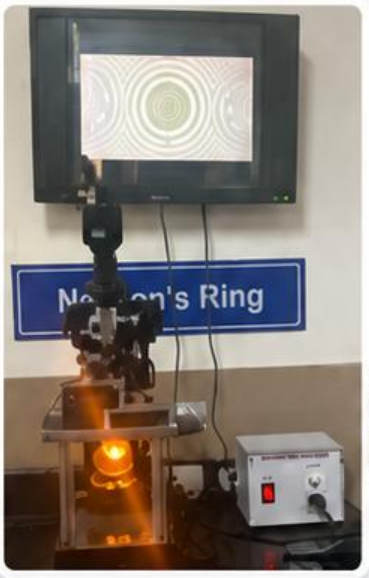
MEDICAPS UNIVERSITY

















APPLIED PHYSICS LAB MANUAL

Faculty of Science
Department of Physics

List of Practical's for B.Tech. First Year Physics Laboratory (common to all branches).

Course Name: Applied Physics (P) **Course Code: PH3001(P)**

Sr. No.	Name of experiment
X.	Introduction to instruments.
1.	To determine the thickness of a wire using a He-Ne laser.
2.	To simulate and analyze a charged particle's spiral trajectory and energy gain in a cyclotron.
3.	To determine the Acceptance angle and Numerical Aperture of an optical fiber by Scanning method.
4.	To determine the wavelength of sodium light by measuring the diameter of Newton's rings.
5.	To determine the wavelength of spectral lines of mercury vapour lamp with the help of Grating and Spectrometer
6.	Determination of Planck's constant (h) using light emitting diode (LED) of various colors.
7.	To determine the frequency of an electrically maintained tuning fork by Melde's experiment.
8.	To study the variation of count rate with applied voltage of a Geiger-Müller (GM) counter, determine its operating voltage, and estimate the efficiency of the GM detector for a radioactive source.
9.	To demonstrate the Heisenberg Uncertainty Principle using single-slit diffraction with a laser.
10.	To study the Hall Effect experiment and find the Hall coefficient and carrier concentration of a given semiconductor crystal.



INTRODUCTION TO INSTRUMENTS.



Experiment No. 0

Objective: Introduction to instruments.

I. SCREW GAUGE

Description

It is based upon the principle of a screw. It consists of a U-shaped metal frame. One end of which carries a fixed stud A whereas the other end B is attached to a cylindrical tube as shown in Fig. 1. A scale graduated in millimetres is marked on the cylindrical tube along its length. It is called Pitch scale.

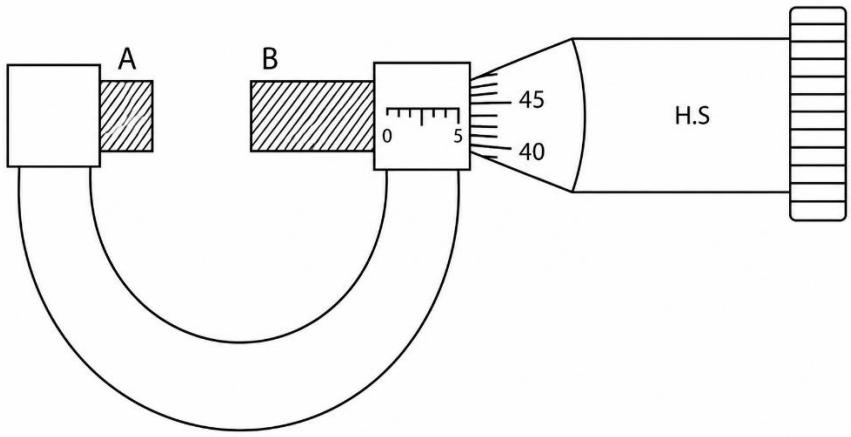


Fig.1 Screw Gauge

The screw carries a head H with a beveled edge. The beveled edge is divided into 100 equal divisions and is called the Head Scale (H.S.). When the head is rotated, the head scale moves along the pitch scale.

Procedure:

1. To find the least count (LC) of the screw gauge

Least count of a screw gauge is the distance through which the screw tip moves when the screw is rotated through one division on the head scale.

$$Pitch = \frac{Distance\ moved\ by\ the\ head\ scale\ on\ the\ pitch\ scale}{Number\ of\ rotations\ given\ to\ the\ head\ scale}$$

$$Least\ count\ (LC) = \frac{Pitch}{Total\ number\ of\ divisions\ on\ head\ scale}$$

To find the pitch, the head or the screw is given say 5 rotations and the distance moved by the head scale on the pitch scale is noted. Then by using the above formula, the least count of the screw gauge is calculated.

Pitch = 5 mm / 5 = 1 mm

Least Count = 1 mm / 100 = 0.01 mm.

The screw head is rotated until the two plane faces A and B are just in contact.

2. **To find the zero correction (ZC)**

i) **Nil error**

If the zero of the head scale coincides with the zero of the pitch scale and also lies on the base line (B.L), the instrument has no zero error and hence there is no zero correction (See Fig).

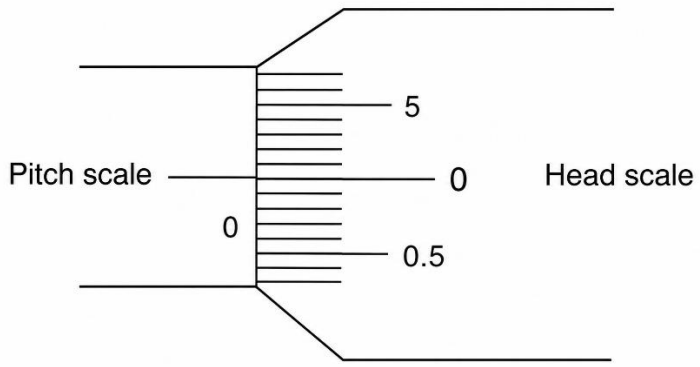


Fig. 2. Nil Error

ii) **Positive zero error**

If the zero of the head scale lies below the base line (B.L) of the pitch scale then the zero error is positive and zero correction is negative. The division on the head scale, which coincides with the base line of pitch scale, is noted. The division multiplied by the least count gives the value of the positive zero error. This error is to be subtracted from the observed reading i.e., the zero correction is negative (See Fig.3).

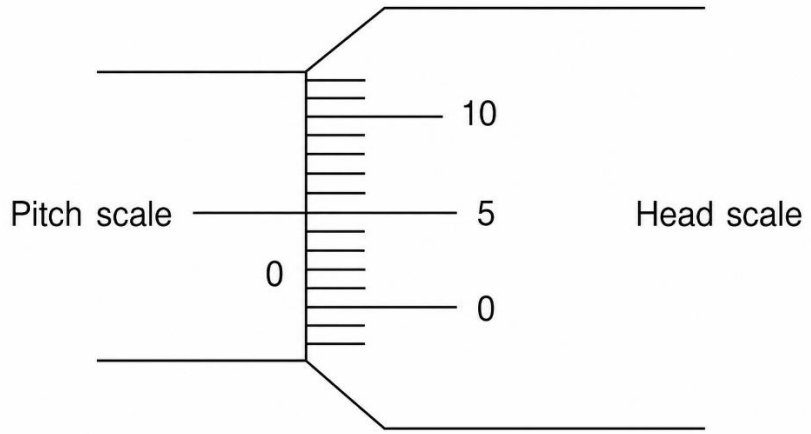


Fig. 3. Positive Zero Error

Example

If 5th division of the head scale coincides with the base line of the pitch scale then

Zero error = +5 divisions

Zero correction = (Z.E x LC) = -(5 x 0.01) = - 0.05 mm.

iii) Negative zero error

If the zero of head scale lies above the base line (B.L) of the pitch scale, then the zero error is negative and zero correction is positive. The division on the head scale which coincides on the base line of pitch scale is noted. This value is subtracted from the total head scale divisions. This division multiplied by the least count gives the value of the negative error. This error is to be added to the observed reading i.e. zero correction is positive (See Fig. 4).

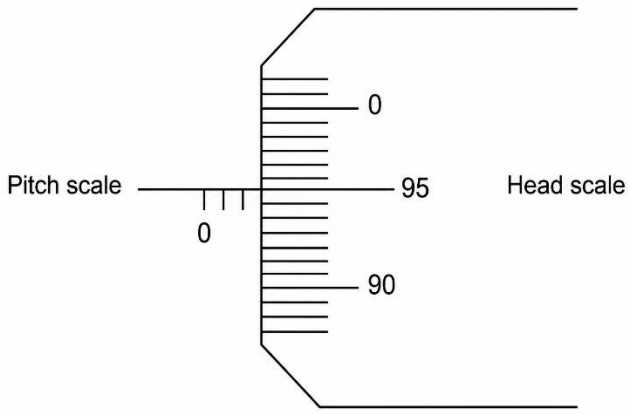


Fig. 4 Negative Zero Error

Example

If the 95th division of the head scale coincides with the base line of the pitch scale then, Zero error = -5 divisions

Zero correction = + 0.05 mm.

3. To find the thickness of the glass plate

The glass plate is gently gripped between the faces A and B. The pitch scale reading and the head scale coincidence are noted. The readings are tabulated.

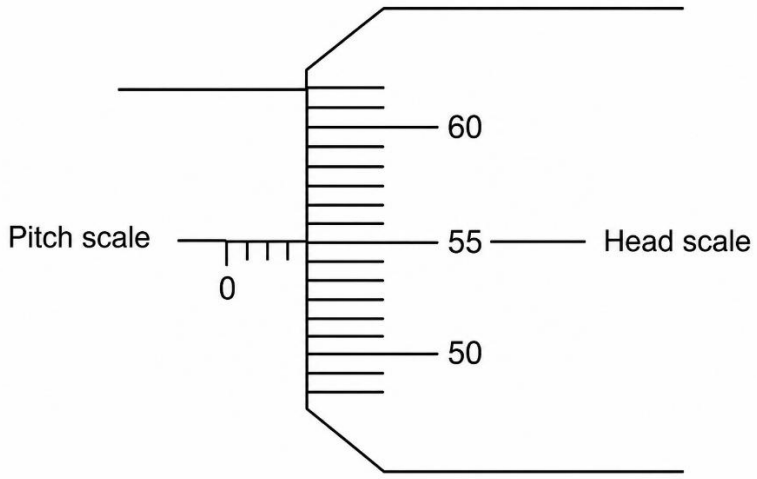


Fig. 5 Screw Gauge Readings

Pitch Scale Reading (P.S.R)

Number of pitch scale division just in front of the head scale fully completed is noted (see Fig. 5). It is measured in millimeter.

Head Scale Coincidence (H.S.C)

Coincidence of head scale division on the base line of the pitch scale is also noted.

Example:

Screw gauge readings: (see Fig. 5)

LC = 0:01 mm Zero error = -3 divisions Zero correction (Z.C.) = +0.03 mm

S.No.	P.S.R mm	H.S.C div	H.S.R = (H.S. C x LC) mm	Total Reading = P.S.R+ H.S.R mm	Corrected Reading = T.R. ± Z.C. mm
1	4	56	0.56	4.56	4.59

II. VERNIER CALIPERS

Description

The vernier calipers consist of a long rigid rectangular steel strip called the main scale (M.S) with a jaw (A) fixed at one end at right angles to its length as shown in Fig.2. The main scale is graduated both in centimeters and inches. The second jaw (B) carrying a vernier scale and capable of moving along the main scale can be fixed to any position by means of a screw cap S. The vernier scale is divided into 10 divisions, which is equivalent to 9 main scale divisions (M.S.D). So the value of 1 vernier scale division is equal to 9/10 M.S.D. The value of 1 M.S.D. is 1 mm

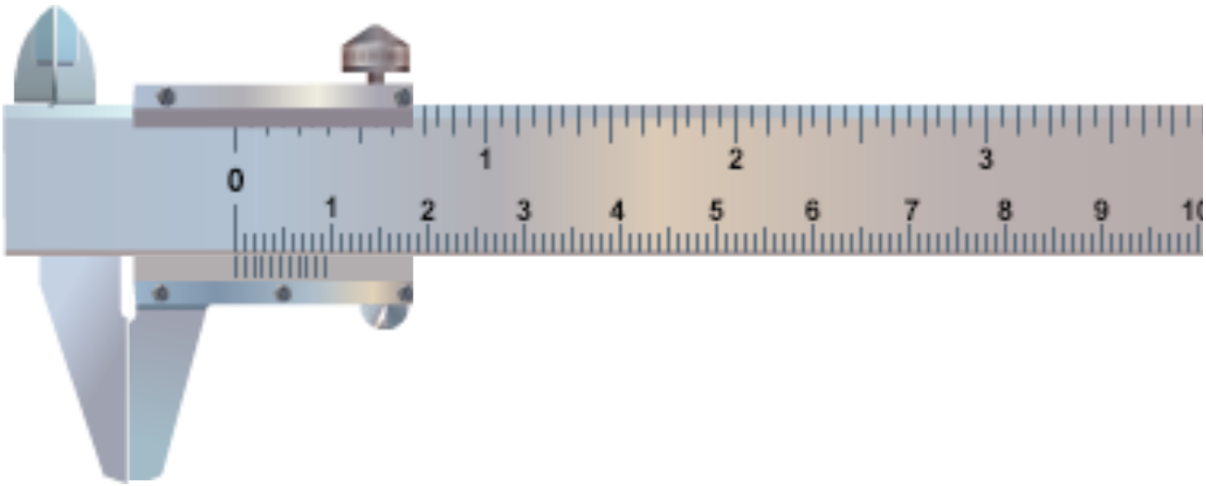


Fig.6. Vernier Calliper.

Procedure:

1. To find the Least Count (LC) of the vernier callipers (see Fig. 7)

It is the smallest length that can be measured accurately by the vernier callipers and is measured as the difference between one main scale division and one vernier scale division.

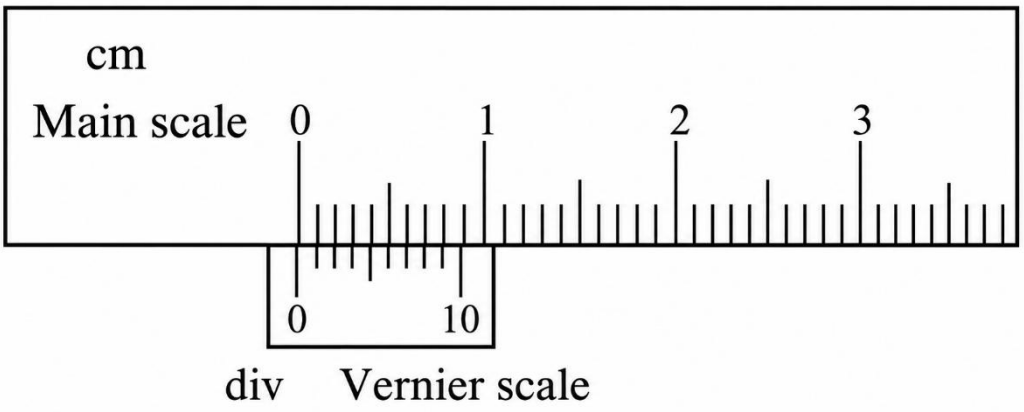


Fig. 7 Vernier scale and main scale.

Least Count (LC) = 1 M.S.D — 1 V.S.D

Value of 1 M.S.D = 1 mm

No of divisions on the vernier scale = 10 divisions.

10 V.S.D =9 M.S.D

1 V.S.D =9/10 M.S.D = 9/10 x 1 mm = 9/10 mm

L.C.=1 M.S.D —1 V.S.D

= 1 mm - 9/10 mm

= 0.1 mm = 0.01 cm

L.C. = 0.01 cm

2. To find the Zero Correction (ZC)

Before taking the readings with the vernier calliper, we must note the zero error of the vernier calliper. When the two jaws of the vernier calliper are pressed together, if the zero of the vernier scale coincides with the zero of the main scale the instrument has no error, otherwise there is a zero error. The zero error is positive if the vernier zero is after the main scale zero. The zero error is negative when the vernier zero is before the main scale zero. Ordinarily, the zero error is

negligible in the case of vernier callipers and so zero error can be considered to be nil.

3. To find the length of the given object

The given object is firmly gripped between the jaws, taking care not to press it too hard. The main scale reading and the vernier coincidence are noted. The main scale reading is the reading on the main scale that is just before the vernier zero. The vernier scale coincidence is found by noting the vernier division that coincides with any one of the main scale. Then the vernier scale reading is found by multiplying the vernier coincidence with the least count. The observations are repeated for various positions of the object.

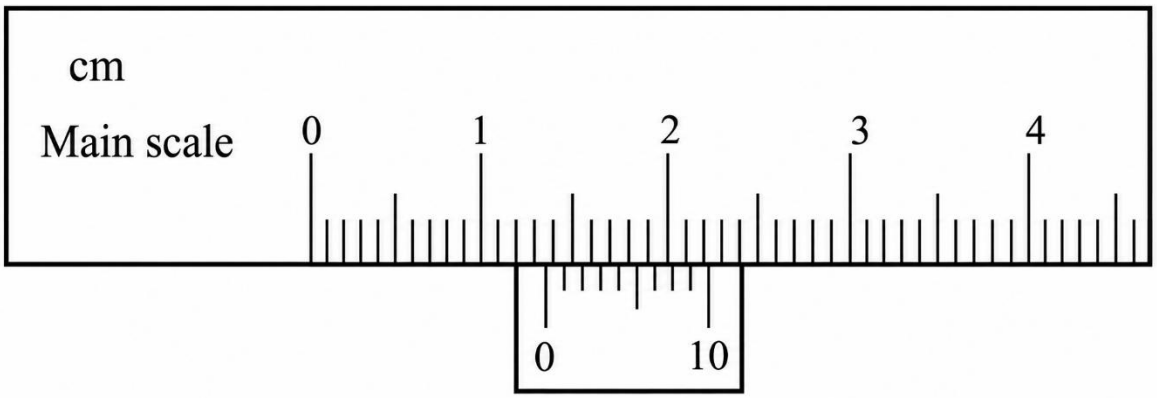


Fig. 8 Vernier Calliper readings.

Example:

Vernier Calliper readings: (See Fig. 8)

LC = 0.01cm

S. No.	M.S.R cm	V.S.C.- div	V.S.R = (V.S.0 x LC) cm	Total Reading = M.S.R+ V.S.R cm	Corrected Reading = T.R. ± Z.C. cm
1	1.3	6	0.06	1.36	1.36

III. THE SPECTROMETER

Description:

i. It has the following three main parts (see Fig.9).

1. The collimator 2. The telescope 3. The prism table.

1. The collimator: Its purpose is to produce a parallel beam of light. It consists of a lens and slit. The slit faces the source of light. The distance between the slit and the lens can be adjusted by a screw fixed to the collimator tube C to obtain parallel beam of rays. The slit consists of two sharp edges. One of the edges is fixed while the other can be moved parallel to it by working the screw provided at its side. The collimator cannot be rotated.



Fig. 9. Spectrometer.

2.The telescope: The telescope T consists of an objective 0 at one and an eyepiece fixed with the cross wires on the other end. The telescope is fixed to the circular scale graduated in degrees. The circular scale is in between the collimator and the telescope. The telescope can be turned with the scale about a vertical axis passing through the center of the spectrometer. The eyepiece of the telescope is provided with cross — wires. The telescope can be fixed in any position by a main screw. Fine adjustment can be made by a fine adjustment screw which is tangential to the main screw. The focusing of the telescope is done by a screw to the side of the telescope.

3. **The prism table:** This consists of a upper plate and a lower plate separated by three springs. Three screws pass through these springs. There are lines engraved in the upper plate so as to mount the prism in proper position. The prism table P can be mounted in any position by means of a screw at its bottom. The prism table can be rotated about the same vertical axis as the telescope.

ii. The initial adjustments:

1. Telescope adjustment: The telescope is turned towards a distant object and its focusing screw is adjusted till the image of the object is clearly seen. In this position, the telescope is capable of receiving parallel rays.

2. Collimator adjustment: The slit is illuminated with sodium vapour lamp or Hg vapour lamp. The telescope is turned so that the telescope and the collimator are in a line. In this position one can see the image of the slit through the telescope. The clear image of the slit is obtained by adjusting the collimator screw. The slit must be adjusted to be narrow and vertical.

3. Levelling of the prism table: This is done with a spirit level. The spirit level is kept on the prism table and the three leveling screws of the prism table are adjusted till the air bubble comes to the centre.

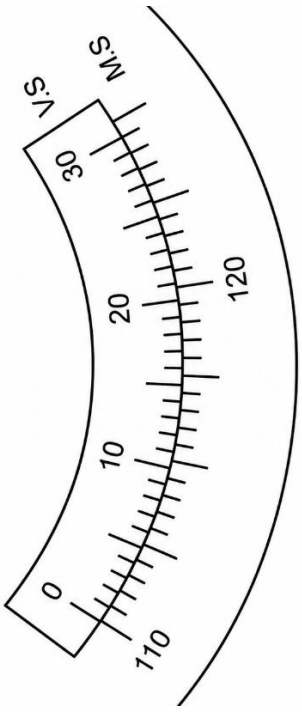


Fig. 10 Circular scale of the spectrometer

Procedure:

1. To find the least count of the spectrometer:

The main scale is a circular scale graduated in degrees. The value of one MSD is half degree. Each vernier scale consists of 30 divisions (Fig. 10).

1 MSD = 0.5° = 30' (10 = 60')

$$LC = \frac{I\ MSD}{Total\ VSD}$$

$$= (30'/30)$$

$$= 1'$$

$$L\ C = 1'$$

2. To read a reading

Main scale reading (MSR) and vernier scale reading (VSR) are noted as explained in the vernier callipers when the telescope and the disc (prism table) are fixed at the required position by the main or fine adjustment screw.

The two vernier scales VA and VB are fixed to the disc, which can be rotated about the same vertical axis. A main screw and a fine adjustment screw as in the telescope control the movement of the vernier scales.

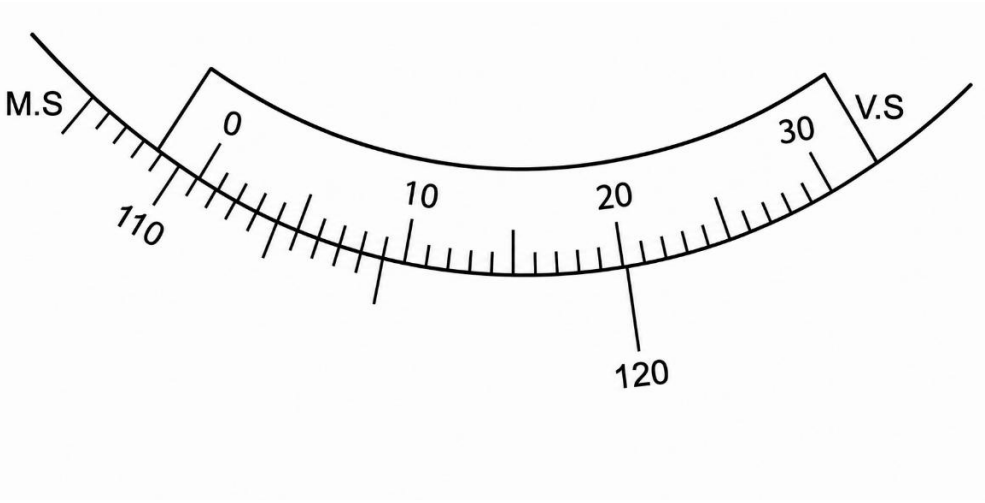


Fig. 11. Spectrometer readings

Example:

Spectrometer readings: (see fig. 11)

LC = 1'

S. No.	M.S.R	V.S.C div	V.S.R = V.S.C x L.C	T.R = M.S.R + V.S.R
1	110.5° = 110° 30'	9	9'	110°39'

THICKNESS OF WIRE

To determine the thickness of a wire using a He-Ne laser.

1.1. Objective: To determine the thickness of a wire using a He-Ne laser.

1.2. Apparatus/Components Required:

i) He-Ne laser with stand
ii) A wire mounted in a holder
iii) Screen/wall
iv) Measuring scale/meter scale
v) Graph paper
vi) Power supply for laser

1.3. Theory: When a monochromatic He–Ne laser beam of wavelength $\boldsymbol{\lambda}$ is incident on a thin wire of thickness $\boldsymbol{d}$, a diffraction pattern is observed on a screen placed at a distance $\boldsymbol{D}$ from the wire. The condition for the first minimum in the diffraction pattern is

$$\boldsymbol{d sin\theta = \lambda}$$

Now, if the half-width of the central maxima is $\boldsymbol{b}$. Assuming the angle θ to be small, we can write, $\boldsymbol{sin\theta \approx \theta \approx b/D}$, which gives the thickness of the wire as,

$$\boldsymbol{d = \frac{D\lambda}{b}}$$

Principle of working: The experiment is based on the phenomenon of Fraunhofer diffraction of monochromatic and coherent light by a thin wire. When a He–Ne laser beam falls on the wire, the light undergoes diffraction at the edges of the wire and produces a series of alternate bright and dark fringes on a screen. The thickness of the wire is determined by measuring the width of the central diffraction maximum.

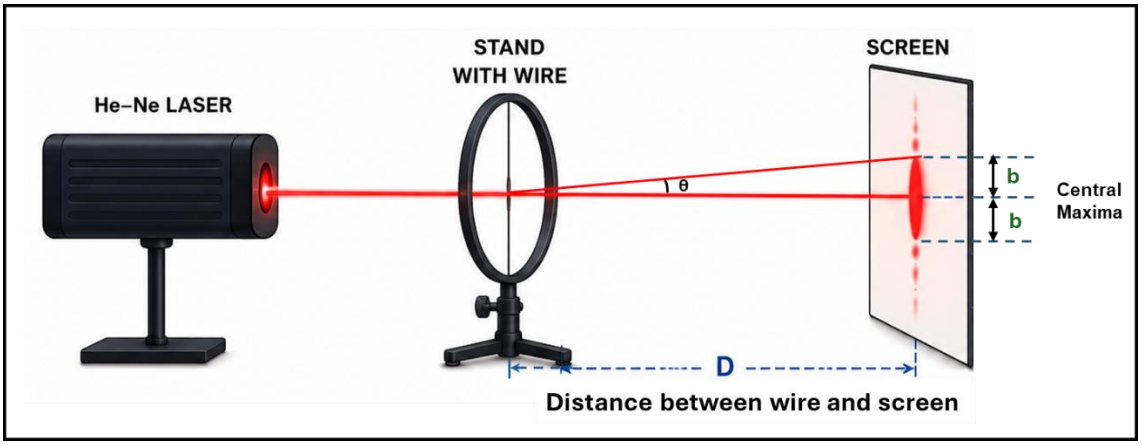


Figure 1.1: Schematic diagram of experimental setup.

1.4. Procedure:

Part A: Alignment

1. Switch on the laser.
2. Allow the laser to stabilize for a short time.
3. Adjust the laser so that the beam travels horizontally and is approximately parallel to the laboratory bench.
4. Mount a wire vertically in the path of the laser beam.
5. Ensure that the laser beam falls approximately on the middle portion of the wire.
6. Place the screen at a sufficiently large distance from the wire, preferably around 1 m or more, so that the diffraction pattern is clearly visible.

Part B: Measurement of the Diffraction Pattern

7. Place a sheet of graph paper on the screen/wall.
8. Observe the diffraction pattern produced by the wire.
9. Identify the central maximum and the first minima on either side.
10. Mark the positions of the two first minima corresponding to the boundaries of the central maximum.

11. Measure the total width (W) of the central maximum.
12. Calculate the half-width: $b = w/2$
13. Measure the distance (D) between the wire and the screen.
14. Calculate the thickness (d) of the wire using

$$d = \frac{D\lambda}{b}$$

15. Repeat the measurements for at least three different values of (D).
16. Calculate the thickness for each observation.
17. Determine the mean thickness of the wire.

Part C: Measurement for the Second Wire

18. Replace the first wire with the second wire without disturbing the laser alignment.
19. Repeat the above procedure for the second wire.
20. Calculate the mean thickness of the second wire.

1.5. Observations

Observation table:

Wavelength of laser λ = 6328 A° (or 6328 x 10^{-7} mm)

For Wire 1

Sr. No.	Distance of the wire from the screen D (mm)	Half width of the Central Maxima b (mm)	The thickness of wire as $d = \frac{D\lambda}{b}$ (mm)	Mean Thickness (mm)
1				
2				
3				

For Wire 2

Sr. No.	Distance of the wire from the screen D (mm)	Half width of the Central Maxima b (mm)	The thickness of slit wire as $d = \frac{D\lambda}{b}$ (mm)	Mean Thickness (mm)
1				
2				
3				

1.6. Calculation:

For wire 1:

d1 = ……

d2 = ……

d3 = ……

Average thickness of wire 1 (d) = ……mm

For wire 2:

d1 = ……

d2 = ……

d3 = ……

Average thickness of wire 2 (d) = ……mm

1.7. Results:

Average thickness of wire 1 = ……mm

Average thickness of wire 2 = ……mm

Percentage error:

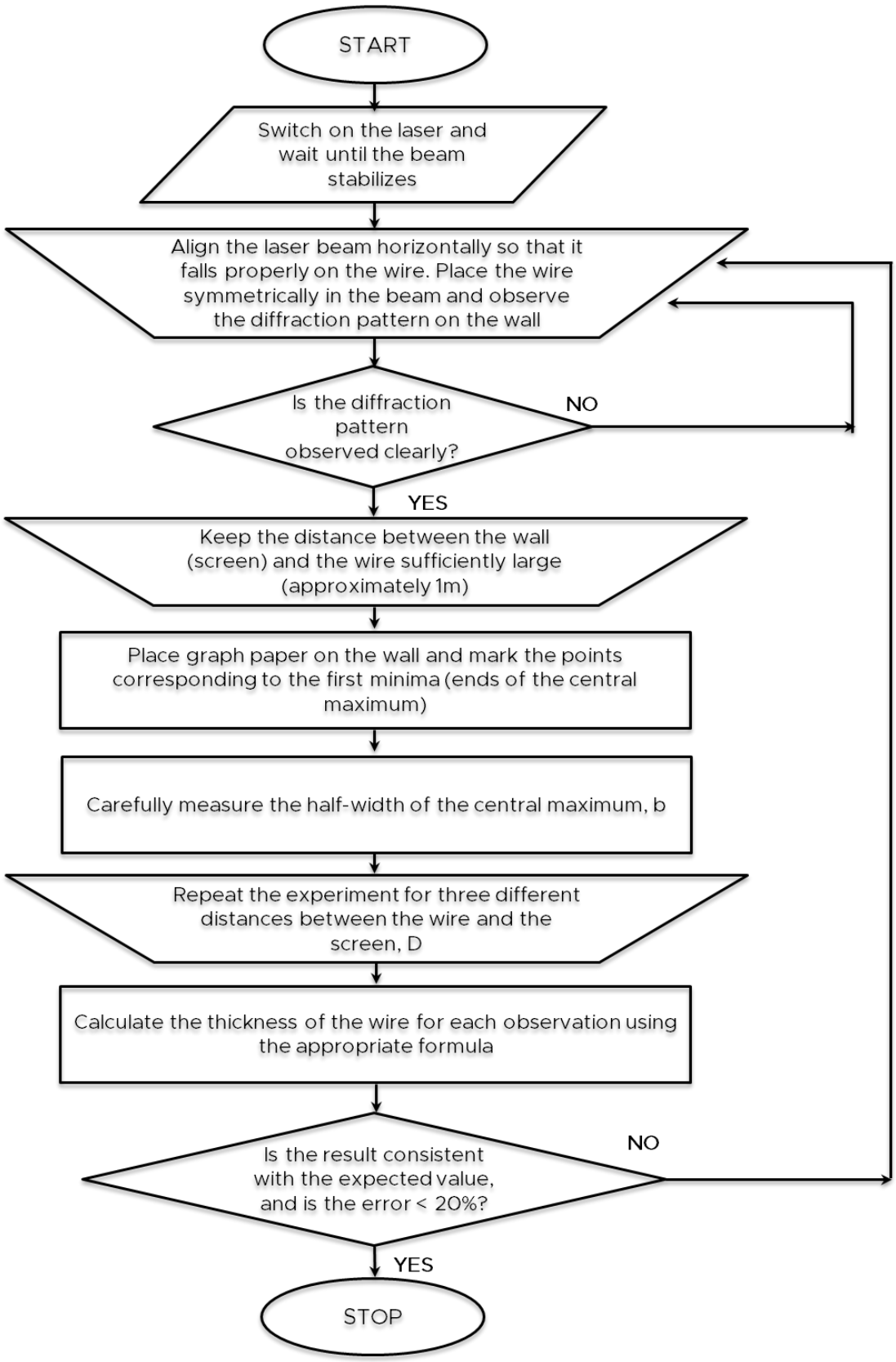
$$\% \text{ error} = \frac{\text{Std. Value} - \text{Expt. Value}}{\text{Std. Value}} \times 100\% = \ldots\ldots\ldots\%$$

Standard values for the given wires are: 0.12 mm & 0.18 mm.

1.8. Precautionary measures to be taken:

1. The laser, wire and screen should be placed on stable supports.
2. The laser beam should fall approximately perpendicular to the wire.
3. The wire should remain fixed throughout the experiment.
4. The laser beam should be properly aligned before taking observations.
5. The screen should be placed sufficiently far from the wire to obtain a clearly measurable diffraction pattern.
6. The positions of the first minima should be identified carefully because they may not appear completely dark.
7. The same reference point should be used while measuring the distance (D).
8. Several readings should be taken at different values of (D) and their mean should be calculated.
9. Avoid vibrations and accidental movement of the experimental setup.
10. Never look directly into the laser beam or its reflected beam.
11. Switch off the laser after completing the experiment.
12. The position of the laser and the wire is not to be disturbed throughout the experiment.

Exp. 1. Flow Chart



1.9. Applications:

The experiment demonstrates the practical application of **laser diffraction for precise measurement of small dimensions**. Its important applications include:

1. **Measurement of thin wires and fibres** – Used to determine the diameter of very thin metallic wires, optical fibres, threads, and other fine objects.
2. **Measurement of small dimensions** – Diffraction techniques can be used to measure dimensions that are difficult to determine accurately using conventional instruments.
3. **Quality control and precision engineering** – Optical diffraction methods are used for non-contact dimensional measurements in manufacturing and precision engineering.
4. **Optical metrology** – The experiment demonstrates the principle of using light as a precise measurement tool, which is important in optical measurement and instrumentation.
5. **Educational application** – It provides a practical demonstration of **Fraunhofer diffraction**, the wave nature of light, and the use of lasers in precision measurements.

1.10. Some sample viva voce questions:

Q1. What is diffraction, and explain its types?

Q2. Why is a laser used in this experiment?

Q3. What is the wavelength of the He–Ne laser used here?

Q5. What are the other methods used to determine the thickness, and how it is different?

Q6. What is the condition for the first minimum?

Q7. Why is the central maximum wider than the other maxima?

Q8. Why is the experiment repeated for different values of (D)?

Q9. What happens to the diffraction pattern when (D) is increased?

Q10. What happens if the wire is made thicker?

Q11. What happens if the wavelength is increased?

Q12. What is the SI unit of thickness?

CYCLOTRON

To simulate and analyze a charged particle's spiral trajectory and energy gain in a cyclotron.

2.1 Objective: To simulate and analyze a charged particle's spiral trajectory and energy gain in a cyclotron.

2.2. Apparatus/Components Required: Personal Computer and Simulation software.

2.2. Theory:

A cyclotron is a machine that accelerates charged particles or ions to high energies. It was invente10.2d to investigate the nuclear structure by E.O Lawrence and M.S Livingston in 1934. Both electric and magnetic fields are used in the cyclotron to increase the energy of the charged particles. As both the fields are perpendicular to each other, they are called cross fields.

In a cyclotron, charged particles accelerate outwards from the centre along a spiral path. These particles are held to a spiral trajectory by a static magnetic field and accelerated by a rapidly varying electric field.

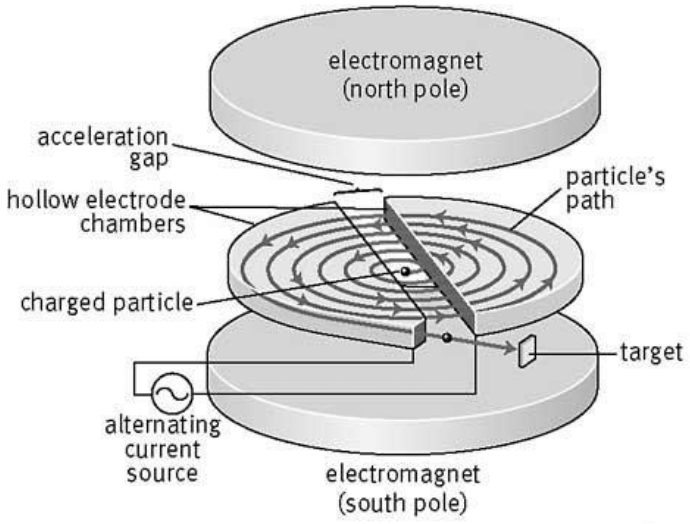


Fig.2.1. Schematic diagram of cyclotron.

Cyclotron and linac are both particle accelerators, but they operate on different principles. Cyclotrons reuse the same acceleration gap multiple times, making them more compact than linacs. Due to their smaller size and simpler design, cyclotrons tend to be less expensive to build and operate compared to linacs. However, the choice between a cyclotron and a linac depends on the specific requirements of the application. Cyclotrons are generally preferred for lower-energy applications, such as isotope production and some types of medical therapy, where compactness, cost-efficiency, and high beam current are essential. Linacs, on the other hand, are better suited for high-energy physics research, cancer therapy requiring high-energy particles, and applications demanding precise beam control.

The cyclotron at TRIUMF is the largest of its kind in the world, measuring almost 18 m (59 feet) in diameter. It produces intense beams of protons of up to 520 MeV.

Working Principle of Cyclotron

- A cyclotron accelerates a charged particle beam using a high frequency alternating voltage which is applied between two hollow "D"-shaped sheet metal electrodes known as the "dees" inside a vacuum chamber.
- The dees are placed face to face with a narrow gap between them, creating a cylindrical space within them for particles to move. Particles are injected into the centre of this space.
- Dees are located between the poles of electromagnet which applies a static magnetic field B perpendicular to the electrode plane.

- The magnetic field causes the path of the particle to bend in a circle due to the Lorentz force perpendicular to their direction of motion.
- An alternating voltage of several thousand volts is applied between the dees. The voltage creates an oscillating electric field in the gap between the dees that accelerates the particles.
- The frequency of the voltage is set so that particles make one circuit during a single cycle of the voltage. To achieve this condition, the frequency must be set to the particle's cyclotron frequency.

The energy of the particles depends on the strength of the magnetic field and the diameter of the dees.

The centripetal force required to keep the particles in a curved path is given by the formula: $$F_c = \frac{mv^2}{R}$$

The force is provided by the Lorentz's force F_B on the magnetic field B

$$F_B = qvB$$

Equating these equations, we get

$$v = \frac{qBR}{m}$$

Hence, the output energy of the particles is given by the expression

$$E_k = \frac{1}{2}mv^2 = \frac{q^2B^2R^2}{2m}$$

Orbital period T is

$$T = \frac{2\pi R}{v} = \frac{2\pi R}{\left(\frac{qBR}{m}\right)} = \frac{2\pi m}{qB}$$

Cyclotron Frequency

$$f = \frac{1}{T} = \frac{qB}{2\,\pi m}$$

A remarkable property of cyclotron frequency is its independence from both the particle's velocity and the radius of its orbit. This means that regardless of how fast a particle is moving or how large its circular path is, as long as the magnetic field strength remains constant, the frequency of its revolution around the field lines remains the same.

For relativistic particle, mass increases $m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} = \gamma \mathrm{m}_o$

therefore, frequency decreases.

The kinetic energy $E_k\ (n)$ grows with every passage of the electric field in the gap between the dees by **q.V**

$$E_k\,(n) = E_k + n\ q.V$$

where E_k is the initial kinetic energy of particle leaving the source i.e. $E_k = \frac{1}{2}mv_0^2$ ($v_0\ is$ velocity of particle leaving the source), n = number of passages through the electric field, V Voltage between dees

The velocity v of the particle when it passages through n number of times in the electric field between the "dees":

$$E_k\,(n) = E_k + n\,q.V$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + n\,q.V$$

$$v = \frac{2}{m}\sqrt{\frac{1}{2}mv_0^2 + n\,q.V}$$

2.3. Procedures:

Turn on the computer and run the simulation. Do as directed in observation copy.

2.4. Observations:

2.4.1 Classical or non-relativistic particle:

Write down the range of

Acceleration voltage (V)……

Magnetic Flux Density B (T) ……

Note: As simulations rely on statistical sampling, it's natural to observe minor differences in outcomes for each run.

a. Set the following values: V = 4000 V and B = 0.1 T, and observe the trajectory of the Proton, Deuteron, and Alpha particle, also note down their kinetic energy (KE).

Trajectory for Proton	Trajectory for Deuteron
K.E. for proton =	K.E. for deuteron =

Trajectory for an alpha particle
K.E. for an alpha particle =

a.Calculate $Vmax$/B ratio (For Proton) (Voltage = 2000 V):

$m_p = 1.67 \times 10^{-27}$ kg
$q_p = 1.6 \times 10^{-19}$ C

Sr. No.	Magnetic Flux Density (T)	$Vmax$ (10^6 m/s)	$Vmax$/B
1.	0.2		
2.	0.3		
3.	0.4		

Comment:

1. At constant voltage, how does the ratio $\boldsymbol{V_{max}}$/B vary with magnetic field B? Write your observation.

2. At constant voltage, how does the R_{max} vary with the magnetic field B? Write your observation.

Calculation for the maximum radius:

$$R_{max} = (mV_{max})/qB$$

b. Note down cyclotron frequency for changing B (For Voltage = 2000 V)

Sr. No.	Magnetic Flux Density (T)	Frequency (MHz)		
		Proton	Deuteron	Alpha
1	0.2			
2	0.3			
3	0.4			

Using the data in the table, determine whether the measured frequencies of the proton, deuteron, and alpha particle at different magnetic flux densities satisfy the relation

$$f = 1/T = qB/2\pi m$$

- Proton ($^1H_1$ nucleus):

$\boldsymbol{m_p \approx 1.67 \times 10^{-27}}$ kg

(For any one observation) $\boldsymbol{f_{Proton}}$ =

- **Deuteron ($^2H_2$ nucleus):**

$\boldsymbol{m_d \approx 3.34 \times 10^{-27}}$ kg

(For any one observation) $\boldsymbol{f_{Deuteron}}$ =

- **Alpha particle ($^4He_2$ nucleus):**

$\boldsymbol{m_\alpha \approx 6.64 \times 10^{-27}}$ kg

(For any one observation) $\boldsymbol{f_{alpha}}$ =

c. Optimizing voltage and magnetic Field to achieve maximum Kinetic Energy

Accelerated Voltage (V)	**Magnetic Flux Density (T)**	**K.E. max**	***n* (number of times particle passage between the "dees")**
2000	0.1		
	0.2		
4000	0.1		
	0.2		

Write your observation.

2.4.2.1 Relativistic without frequency adjustment

For voltage = 100000 V and B (T) = 2 T

Particle	Max velocity attained (10^6 m/s)	Frequency (MHz)
Proton		
Deuteron		
Alpha		

Write your observation.

2.4.2.2 Relativistic with frequency adjustment

For voltage = 100000 V and B (T) = 2 T

Particle	Max Velocity attained (10^6 m/s)	Frequency (MHz)
Proton		
Deuteron		
Alpha		

Write your observation.

2.5 Results:

1. The trajectory of the particles within the cyclotron can be visualized graphically.
2. Study how changes in magnetic field, electric field, and other parameters affect the results.
3. Verify the simulation outcomes with the expected behavior based on cyclotron theory.
4. Adjust parameter to get better cyclotron's performance.
5. Understand the limitations of Cyclotron for smaller charged particle and higher energy particles.

2.5 Precautionary Measure to be taken:

Careful handling of computer and simulation Program.

2.6. Some Sample questions:

1. What role do magnetic and electric fields play in operation of cyclotron?
2. Why is a cyclotron not suitable for accelerating electrons? What limitations does it have in accelerating heavy particles?

3. How does the frequency of the alternating voltage applied to the dees relate to the motion of the particle?
4. Discuss the concept of resonance in a cyclotron. Why is it essential for the efficient acceleration of particles?
5. Explain the relativistic limitations of a cyclotron. How do synchrotrons overcome these limitations?
6. What are the factors that determine the maximum energy attainable by a particle in a cyclotron?
7. How does the design of a cyclotron differ for accelerating protons and heavy ions?
8. Discuss the applications of cyclotrons in medicine, research, and industry.
9. A cyclotron is designed to accelerate protons. If the magnetic field is doubled, how does the cyclotron frequency change?
10. If the radius of the dees in a cyclotron is increased, how does it affect the maximum energy of the accelerated particles?
11. A cyclotron is used to accelerate alpha particles. How would you modify the cyclotron to accelerate protons without changing the magnetic field?
12. A charged particle is injected into a cyclotron at an angle to the magnetic field. Describe the path of the particle.
13. A cyclotron is operating at a fixed frequency. How can the energy of the accelerated particles be increased without changing the frequency?

NUMERICAL APERTURE OF AN OPTICAL

To determine the Acceptance angle and Numerical Aperture of an optical fiber by Scanning method.

3.1. Objective: To determine the Acceptance angle and Numerical Aperture of an optical fiber by Scanning method.

3.2. Apparatus/Components Required:

1. Bread Board
2. Optical Fiber
3. Optical Chunks
4. Laser Diode
5. Lens
6. Photodiode

3.3. Theory: Numerical aperture (N. A.) is a basic descriptive characteristic of specific fiber. It can be thought of as representing the size of "degree of openness" of the input acceptance cone (Fig 3.1) Mathematically numerical aperture is defined as sine of acceptance angle

$$\text{N. A.} = \text{Sin}\theta_a$$

The light gathering power or flux carrying capacity of fiber is numerically equal to the sequence of the aperture, which is the ratio between the area of a unit sphere with the acceptance cone and the area of the hemisphere (2θ solid angle).

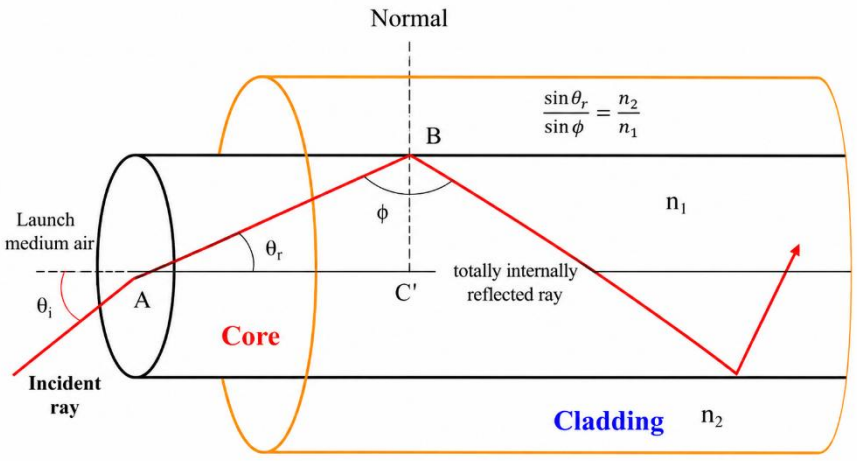


Figure 3.1: Ray propagation in fiber

Snell's law can be used to calculate the maximum angle within which light will be accepted into and conducted through a fiber.

$$N.A. = \sin\theta_a = \sqrt{n_1^2 - n_2^2}$$

Where $\sin\theta_a$ is numerical aperture, n_1 and n_2 are the refractive indices of the core and the cladding. The angle θ_a of the step index fiber is determined by the critical angle $\theta_c = Sin^{-1}(n_2/n_1)$ for total internal reflection to take place at core cladding interface. For a ray with incident angle $\theta_i > \theta_a$, the ray undergoes only partial refection at core-cladding interface and is radiate out into the cladding.

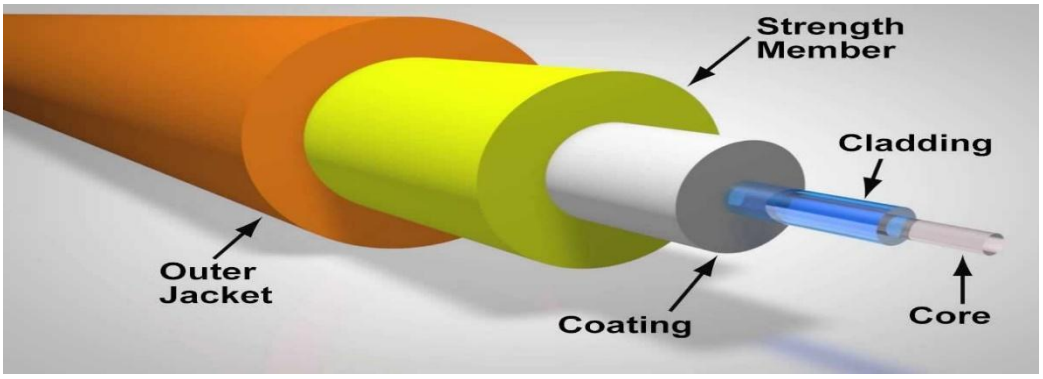


Figure 3.2: Structure of Optical Fiber.

In short length of straight fiber, ideally a ray launched at an angle θ, at the input end should come out at the same angle θ from output end. Therefore, the far field at the output end will also appear as a cone of semi angle θ_a emanating from the fiber end. It is then simpler to make measurements on this far field to determine the N. A. of the fiber.

3.4. Procedure:

1. Identify all the components.
2. Mount both the ends of the optical fiber on the fiber optic chucks.
3. The 10X microscopic objective is used to effective coupling of light onto one of the fiber ends using a 10X microscopic objective.
4. Couple the light into the fiber to obtain maximum power.
5. Note the distance between the output fiber tip and the detector. Let it be 'r'.
6. Place the detector at some distance from the output end (other than at which light is coupled) of the fiber such that it is perpendicular to the axis of the fiber.
7. Move the detector on the extreme left or right to obtain minimum power.
8. Now move the detector in step of 0.5 mm (x) and note the corresponding output power.
9. Proceed the above step again till the min. power is reached

10. Use the formula to obtain θ= $\tan^{-1}(x/r)$ for different value of x & r.
11. Plot a graph between θ and output power P (μW) taking θ on x axis and power on y axis.
12. Calculate 5% of max. power from the graph paper,
13. Draw a line parallel to x axis at 5% of max. Power.
14. Drop two perpendiculars from point of intersection on x axis
15. Difference of these two readings on x axis gives 2θa
16. Acceptance angle calculated by the formula

$$\theta_a = \frac{|\theta_2 - \theta_1|}{2}$$

3.5. Observations:

Least count of the scale graduated on the translational mount = ……. mm

Distance between the detector and output end of the fiber (r) = …….mm

Observation Table:

Sr. No.	x (mm)	Power (μW)	$\frac{x}{r}$	$\theta = \tan^{-1}\left[\frac{x}{r}\right]$		Sr. No.	x (mm)	Power (μW)	$\frac{x}{r}$	$\theta = \tan^{-1}\left[\frac{x}{r}\right]$
1						15				
2						16				
3						17				
4						18				
5						19				
6						20				
7						21				
8						22				
9						23				
10						24				
11						25				
12						26				
13						27				
14						28				

3.6. Calculation:

Plot a graph between angle (θ) and power. Draw a horizontal line parallel to x-axis at 5% of the maximum power and find θ_1 and θ_2 by dropping perpendiculars on x-axis.
Acceptance angle

$$\theta_a = \frac{|\theta_2 - \theta_1|}{2}$$

Acceptance Cone = $2\theta_a$

N.A. = $\sin\theta_a$

3.7. Results:

Acceptance angle of optical Fiber=……..

Numerical Aperture of optical Fiber = ……..

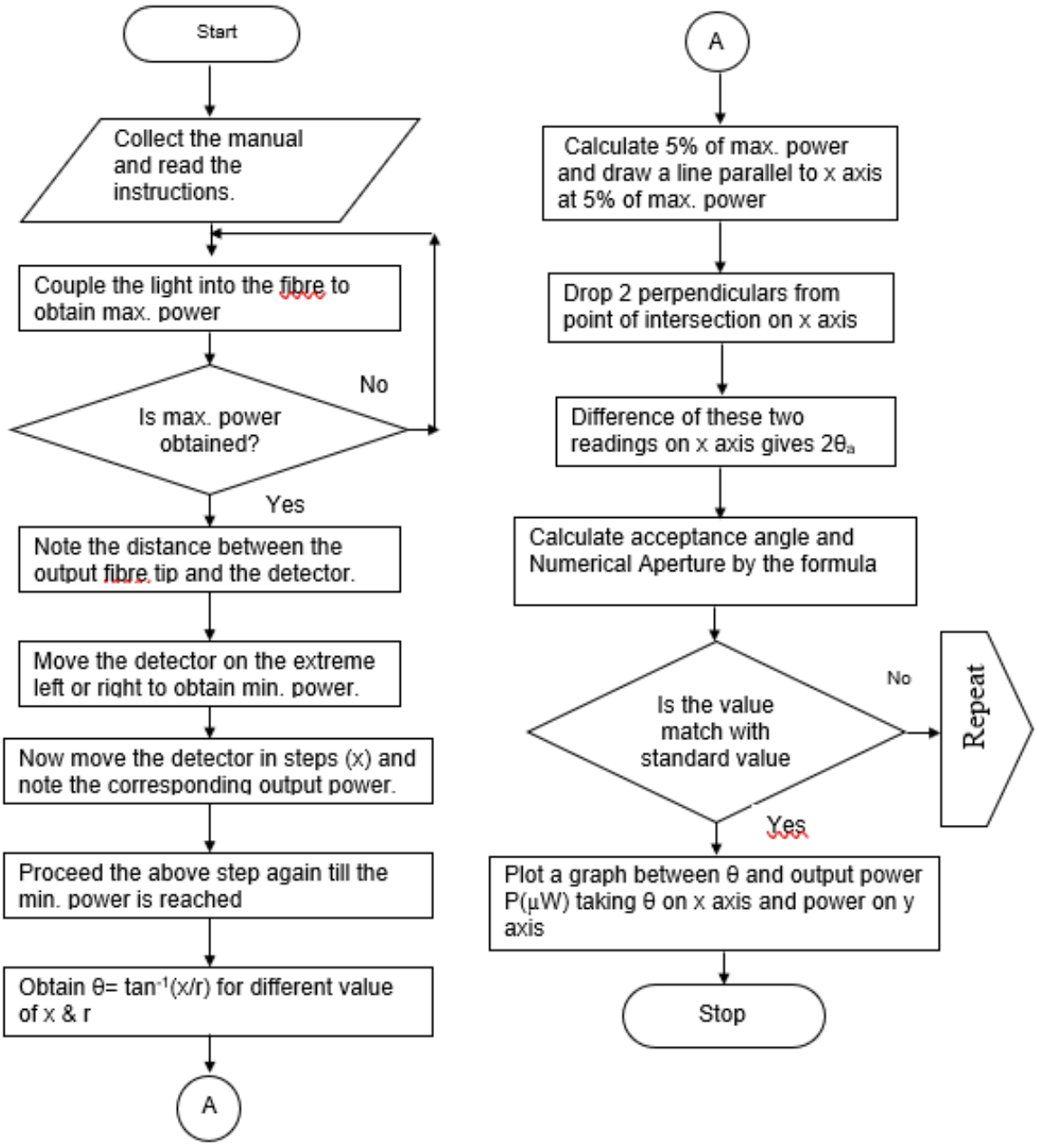
Percentage error:

$$\% \text{ error} = \frac{\text{Std Value} - \text{Expt Value}}{\text{StdValue}} \times 100 =$$

3.8. Precautionary measure to be taken:

1. Coupling of LASER light to the fiber should be very proper.
2. Fiber should be kept straight.
3. Light should fall normally on the fiber.
4. Observation of power meter should be measured accurately.
5. Direct viewing of LASER light should be avoided.

Exp. 3 NA Flow Chart



3.9. Applications of Planck's Constant experiment

- Determining the acceptance cone of a fiber
- Measuring the light-gathering capability of the fiber
- Comparing NA of different fiber
- Studying how bending, wavelength, or core/cladding structure affects the angular output distribution
- Characterizing multimode and specialty fiber

3.10. Some sample viva-voice questions:

1. What is Acceptance angle and Acceptance cone?
2. How Total internal reflection can be achieved in Optical Fiber?
3. What are the main advantages of using optical fibers over other transmission mediums?
4. What are the main parts of an optical fiber?
5. What are guided modes in optical fiber?

NEWTON'S RING

To determine the wavelength of sodium light by measuring the diameter of Newton's rings

4.1. Objective:

To determine the wavelength of sodium light by measuring the diameter of Newton's rings.

4.2. Apparatus/Components Required:

1. Sodium lamp with transformer
2. Traveling microscope

Newton's ring apparatus consists of an optically plane glass plate, convex lens, and a glass plate movable at angle 45^0.

4.3. Theory:

Principal of working:

The experimental arrangement of obtaining Newton's ring is shown in Figure 4.1 (b).

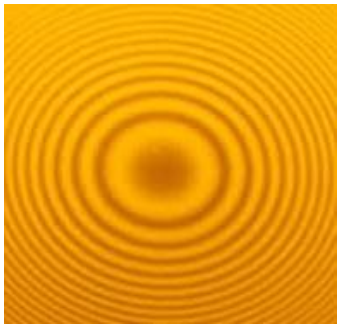


Figure 4.1(a): Newton's ring

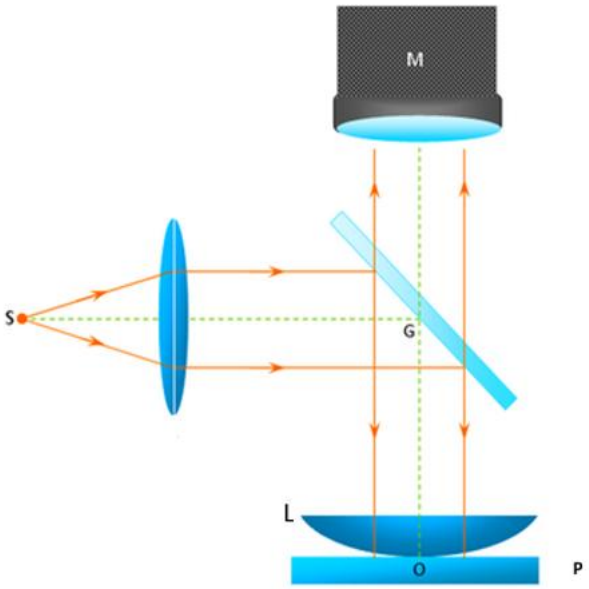


Figure 4.1(b): Newton's ring experimental setup

L is a Plano convex lens of large radius of curvature. This lens with its convex surface is placed on a plane glass plate P. The lens makes contact with the plate at O. Light from an extended monochromatic source falls on the

glass plated G kept at 45^0 with the vertical. The glass plate G reflects a part of light normally on the air film enclosed by the lens L and the glass plate P. Newton's rings are observed due to the interference of two rays one which is reflected from the top of the Plano convex lens and other ray which is reflected from the bottom of the Plano convex lens. These rings are localized in the air film and can be seen with the help of microscope focused on the film.

The formation of Newton's rings can be explained with the help of Figure 4.2.

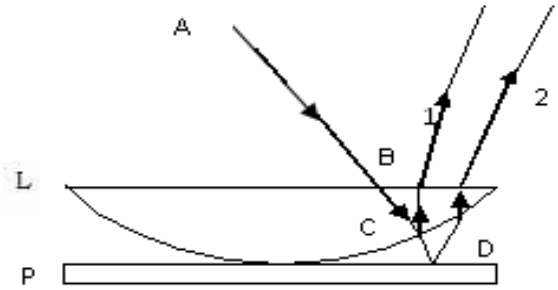


Figure 4.2

AB is a monochromatic ray of light which falls normally on the plano convex lens. A part of incident ray is reflected from point C which goes out in the form of ray 1 while the other part of the ray is refracted along CD, after reflection from D it comes out as ray 2, with a phase reversal of ray 1 and 2 satisfies the condition of interference and so they interfere and produces dark and bright fringes. As the rings are observed in the reflected light, the path difference between them is $2\mu t \cos r + \lambda/2$.

For air film $\mu = 1$, and for normal incidence $r = 0$ hence in the case, the path difference is $2t + \lambda/2$. At the point of contact $t = 0$, and the path difference is

λ/2, which is the condition of minimum intensity. Thus, the central spot is dark. For n^{th} maximum we have $2t + \lambda/2 = n\lambda$.

Newton's ring by reflected light

Diameters of dark and bright rings can be calculated as follows: In fig. 4.3 let LOL' be the lens placed on a glass plate AB. The curved surface LOL' is the part of spherical surface with center at C. let R be the radius of curvature and r be the radius of Newton's ring corresponding to the constant film thickness t. as discussed above.

Substituting the values $r \times r = t \times (2R - t) = 2Rt - t^2 >> 2Rt$

$$r^2 = 2Rt \quad \text{or } t = r^2/2R$$

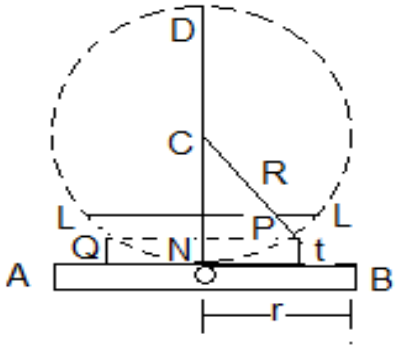


Figure 4.3

Thus, for bright ring $r^2 = [(2n - 1)\lambda R]/2$

Replacing r by D/2 we get the diameter of nth bright ring as

$$\frac{D^2}{4} = \frac{(2n-1)\lambda R}{2}$$

$$D = \sqrt{2\lambda R}\sqrt{2n-1}$$

Or $$D \propto \sqrt{2n-1}$$

i.e. the diameters of bright rings are proportional to the square root of odd natural numbers.

Similarly, for dark ring

$$2\frac{r^2}{2R} = n\lambda$$

$$r^2 = n\lambda R$$

Or $$D = 2\sqrt{n\lambda R}$$

$$D \propto \sqrt{n}$$

Thus, the diameter of dark ring is proportional to square root of natural numbers.

Now as we have

$$D_n^2 = 4n\lambda R \quad \text{and} \quad D_{n+p}^2 = 4(n+p)\lambda R$$

$$D_{n+p}^2 - D_n^2 = 4p\lambda R$$

Or

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR}$$

Using this formula Wavelength of sodium light.

4.4. Procedure:

1. First the glass plate P, Plano convex lens L and plate P are cleaned to ensure that there are no dust particles.
2. Now the light from lens L falls on the glass plate G which is to be kept at 45^0. This ensures that light is falling normally on the system of Plano convex lens and plane glass plate in contact beneath the glass plate G.
3. Note down the least count of traveling microscope.

4. Move the eye piece to and fro inside the tube so as to adjust it such that cross wires are distinctly seen.
5. Focus the microscope to obtain distinct Newton's rings in the field of view by lowering or raising the microscope by rack and pinion arrangement.
6. The microscope is moved horizontally with the help of tangential screw and counting the rings, the cross wire is made tangential on the 12th bright circular ring. The reading of microscope is noted. Now the microscope is gradually moved towards left and the readings of 12th, 10th, 8th, and 6th rings are noted. Since it is difficult to adjust the cross wire tangential to the rings nearer to center hence the readings are taken only up to 6th ring.
7. Now keep on moving the tangential screw of the microscope in the same direction till the 6th ring on the right-hand side is reached. Again, note the readings of 6th, 8th, 10th, and 12th rings. The difference in readings on the two sides of a ring gives its diameter.

4.5. Observations:

For the determination of the diameter of Newton's Ring $p = 2$ and $R = 550\ mm$.

$$\text{Least count of the apparatus} = \frac{\text{Pitch}}{\text{Total no. of vernier scale divisions}} = \quad \text{mm}$$

Observation table for the diameter of rings: -

S. No	No of Rings	Readings of microscope						Diameter of Rings D_n = a- b	D_n^2	$D_{n+p}^2 - D_n^2$
		On the left side			On the right side					
		MSR	VSR	Total Reading (a)	MSR	VSR	Total Reading (b)			
1	12th									$D_{12}^2 - D_{10}^2$=
2	10th									

3	**8th**									$D_{10}^2 - D_8^2$=
4	**6th**									$D_8^2 - D_6^2$=

4.6. Calculation:

The radius of curvature of Plano convex lens is = 550 mm.

Wavelength of sodium light can be calculated by using the formula

$$\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR} A^0$$

4.6. Results: The wavelength of sodium light = …….. A^0.

The standard Wavelength of sodium light = 5893 A^0

Percentage error:

$$\% \text{ error} = \frac{\text{Std Value} - \text{Expt Value}}{\text{StdValue}} \times 100 =$$

4.7. Result:

- The wavelength of Sodium light =…….. Å

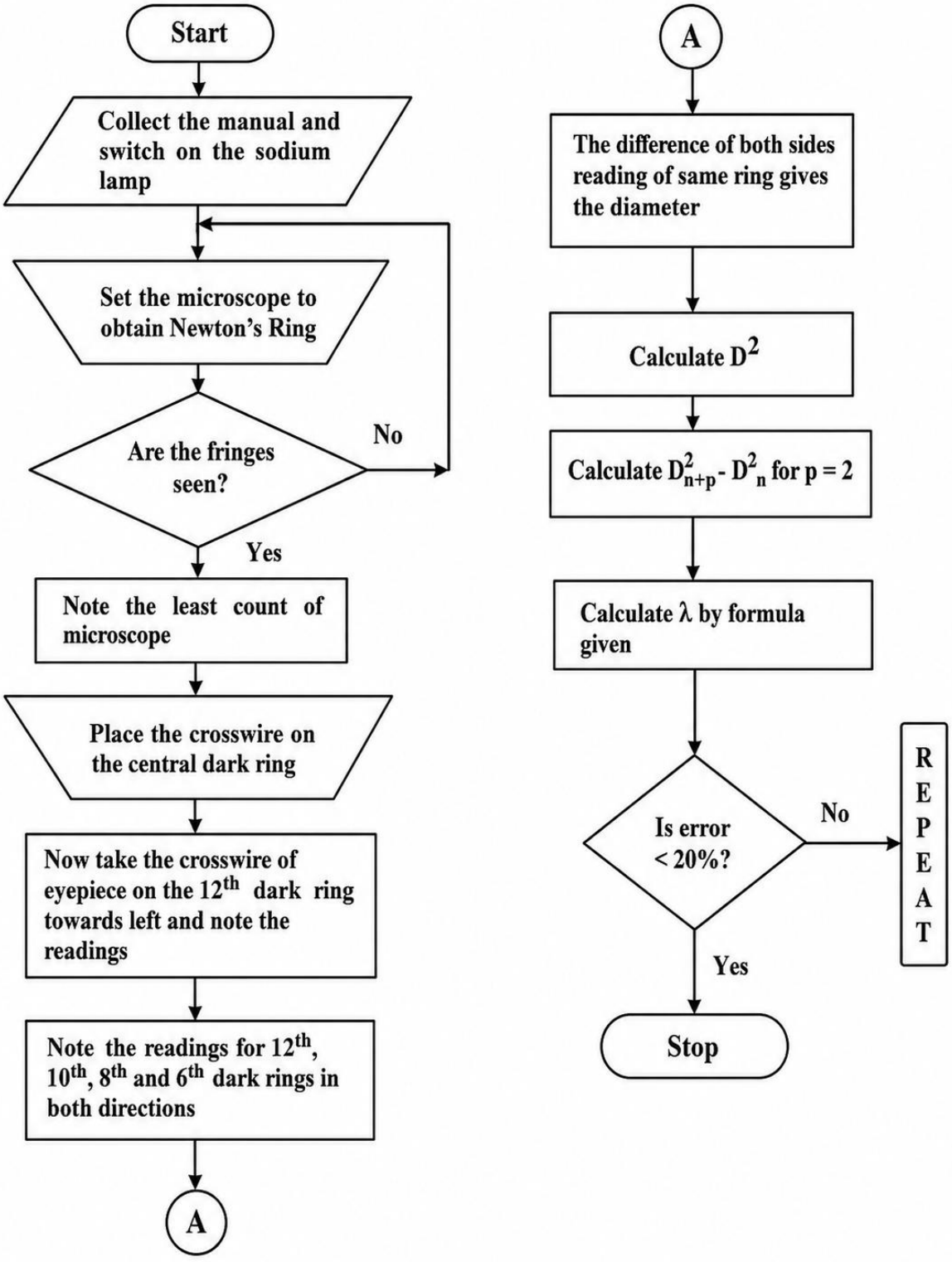
4.8. Precautionary measure to be taken:

1. The glass plate and the lens must be perfectly clean.
2. The glass plate should be kept at 45^0.
3. The radius of curvature of the Plano convex lens must be large so that the large diameter rings may be obtained.
4. The source must be monochromatic and extended.
5. The tangential screw of the microscope must be moved in the same

direction to avoid back lash error.

6. While measuring the diameter the cross wire should be tangential to the bright ring.

Exp. 4. Newton's ring Flow Chart



4.9. Applications of Newton's Ring experiment

1. **Determination of the wavelength of monochromatic light**
   - The most common laboratory application.
   - The wavelength is determined by measuring the diameters of successive dark or bright rings.
2. **Measurement of the radius of curvature of a plano-convex lens**
   - If the wavelength is known, the radius of curvature of the lens can be calculated accurately.
   - Used in optical laboratories for lens characterization.
3. **Determination of the refractive index of liquids**
   - When a liquid is introduced between the lens and glass plate, the ring diameters change.
   - The refractive index is obtained by comparing ring diameters in air and in the liquid.

4.10. Some sample viva-voice questions:

i. What is the basic principle of Newton's rings experiment?
ii. Why are we getting circular ring pattern in Newton's experiment rather than parallel fringes?
iii. Why is the center of the rings observed as dark in Newton's rings?
iv. What happens if we replace sodium light with white light? Can we determine the wavelength of white light using Newton's rings experiment?
v. If we replace yellow light with green light, is there any difference in the formation of rings?
vi. What is the purpose of glass plate incline in this experiment?
vii. Why it is necessary for the light to fall normally on Plano convex lens?
viii. What is the purpose of glass plate incline at in this experiment?
ix. What are the applications of Newton's ring experiment?

DIFFRACTION GRATING

To determine the wavelength of spectral lines of mercury vapour lamp with the help of Grating and Spectrometer

5.1. Objective: To determine the wavelength of spectral lines of mercury vapour lamp with the help of Grating and Spectrometer

5.2. Apparatus/Components Required: Spectrometer, grating, mercury lamp, power supply, magnifying glass, and table lamp.

5.3. Theory: Diffraction and Fraunhofer Diffraction: The spreading-out (bending or deviation from its original path) of a wave when it passes through a narrow opening is usually referred to as diffraction and the intensity distribution on the screen known as the diffraction pattern.

In the Fraunhofer diffraction, the source and the screen are at infinite distance from the aperture or slit, this is easily achievable by placing the source on the focal plane of a convex lens and placing the screen on the focal plane of another convex lens

The two lenses effectively moved the source and screen to infinity because the first lens makes the light beam parallel and second lens effectively makes the screen receives a parallel beam of light. Further, the Fraunhofer diffraction pattern is not difficult to observe; all that one needs is laboratory spectrometer. The collimator renders a parallel beam of light, and the telescope receives parallel beam of light on its focal plane. When the value of N is very large, one obtains intense maxima, when

$$dsin\theta = n\lambda \quad \textbf{(n=0, 1, 2)}$$

Components of Spectrometer:

Diffraction Grating:

An arrangement which essentially consists of many equidistant slits is known as a diffraction grating; the corresponding diffraction pattern is known as the grating spectrum. The principal maxima corresponding to different spectral lines will

correspond to different angles of diffraction. Thus, the grating spectrum provides us with an easily obtainable experimental set up for determination of wavelengths.

Spectrometer:

The spectrometer is an instrument which is usually employed for the study of the composition of light and the measurement of refractive indices of solids and liquids. It consists essentially of three parts.

(i) Prism table
(ii) collimator and
(iii) telescope.

Adjustment: Before using the spectrometer, the following adjustments must be made.

A. The axis of the telescope and that of the collimator must intersect the principal vertical axis of rotation of the telescope.

This adjustment is already being done by the manufacturer and can only be tested in the laboratory. For this purpose, a pin is mounted vertically in the center of prism table, and its image is observed in the telescope tube without eyepiece and for a wide slit in the collimator. If the image appears in the middle, at every position of telescope then the adjustment is perfect. Otherwise, the image is made in the center by using the screws supporting the telescope and collimator till the pins appear in the middle.

B. Prism table should be leveled

1. The prism table is leveled with help of three screws supporting the prism table. This is done with the help of spirit level.
2. The second method which is generally used is optical leveling of the prism table.

C. Telescope and collimator are adjusted for parallel light by Schuster's method.

First, prism is placed on the prism table and then adjusted approximately for minimum deviation position (see procedure). The spectrum is now seen through the telescope. Now the refracting edge of the prism is slightly turned towards the collimator, and the collimator is adjusted with the help of its rack and pinion arrangement such that spectrum is distinctly seen. Then the refracting edge of the prism is slightly turned towards the telescope, and telescope is adjusted with the help of its rack and pinion arrangement such that again the spectrum is distinctly seen. The process is repeated till the slight rotation of the prism table does not make the image go out of focus. This means that both collimator and telescope are now individually set for parallel rays.

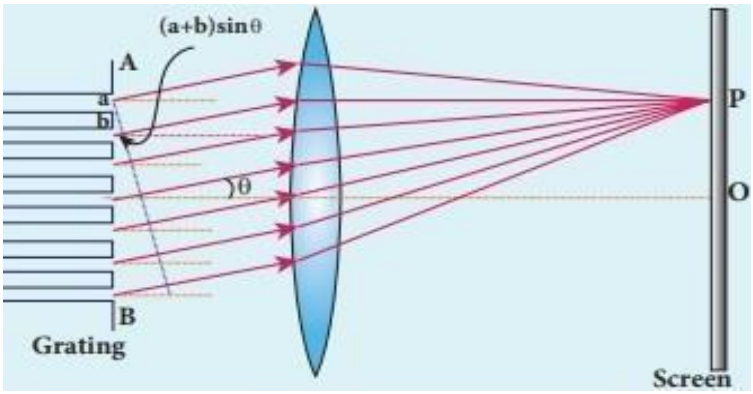


Figure 5.1(a): Diffraction of a plane wave incident normally on a multiple slit.

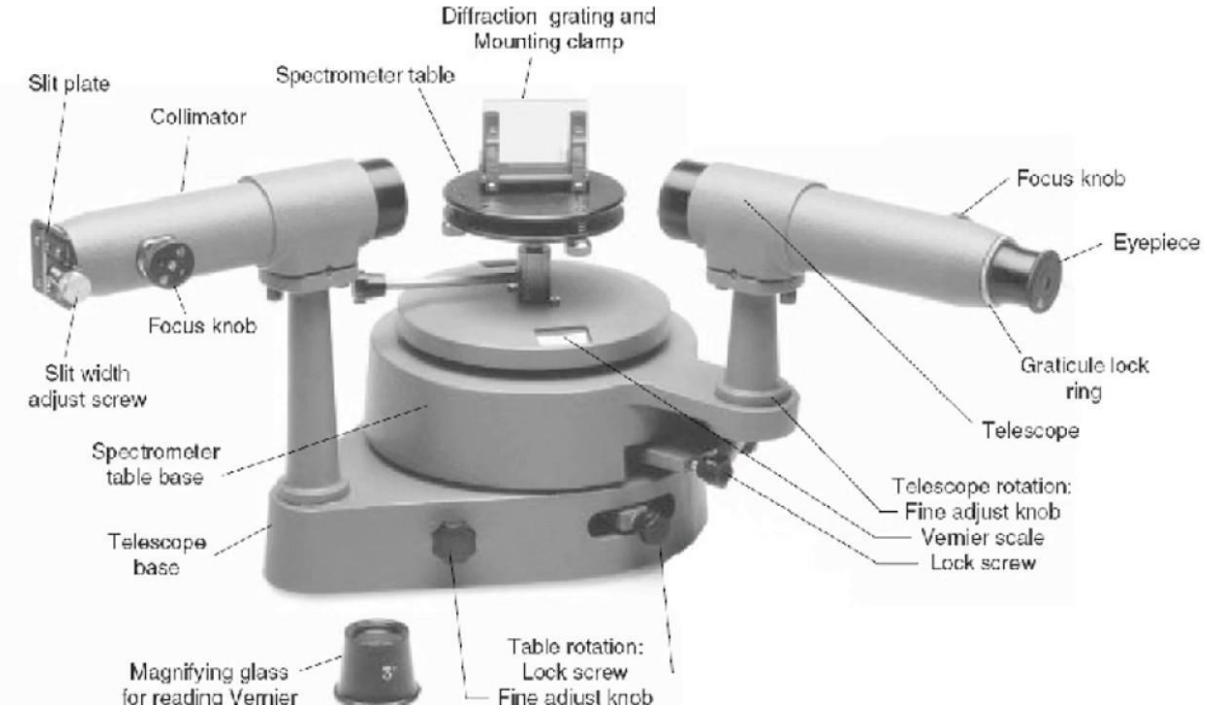


Figure 5.1 (b): Grating Spectrometer

5.4. Procedure:

1. Before the actual measurement of the angle of diffraction, the adjustment of spectrometer is made as given above.
2. Collimator and telescope are arranged in a line such that direct image of slit is visible. Coincide with the vertical cross wire on the direct image of slit and note the reading of the vernier scale.
3. Rotate the telescope by 90^0.
4. Rotate the grating table such that the slit image is visible in the telescope. coincide the vertical cross wire on the image and note the reading of vernier.
5. Turn the prism table by 45^0 or 135^0. In this situation grating is normal to the incident beam.

6. Slit should be adjusted parallel to the lines of the grating. For this setting, the slit is rotated in its own plane till the spectral lines become very sharp and bright.
7. Rotate the telescope to the left side of direct beam image and see the 1st order spectrum. Adjust and superimpose the vertical cross wire on different spectral lines (i.e. red, yellow and violet). Note the reading of Vernier for different spectral lines.
8. Find the diffraction angle for colours.

5.5. Observations

Observation table:

For the determination of the angle of diffraction

Number of lines on the grating (N) = 15000 lines/inch.

Value of the one division on the main scale of spectrometer = …….. Total divisions on the Vernier scale of spectrometer =……

$$Least\ count\ = \frac{Value\ of\ the\ 1\ division\ on\ the\ main\ scale}{Total\ number\ of\ divisions\ on\ the\ vernier\ scale} = _____$$

Order of the spectrum (n)	Colour of the line	Spectrum on the left of direct image (in degree)			Spectrum on the right of direct image (in degree)			Difference of the two readings of the same (2θ) (a-b)	Avg. Value of (2θ)	Diffraction angle for colour (θ)
		M S	V S	**Total (a)**	M S	V S	**Total (b)**			
n = 1	Violet V1									θ_V
	V2									
	Red V1									θ_R
	V2									

Standard wavelength values of colors

Violet - 4046 Å, Red - 6800 Å

5.6. Calculation:

$$esin\theta = n\lambda$$

here, *e* is the grating element = $\frac{2.54}{N}$, where N is the number of lines on the grating (N).

5.7. Results:

- The wavelength of violet line =..... Å
- The wavelength of red line =..... Å

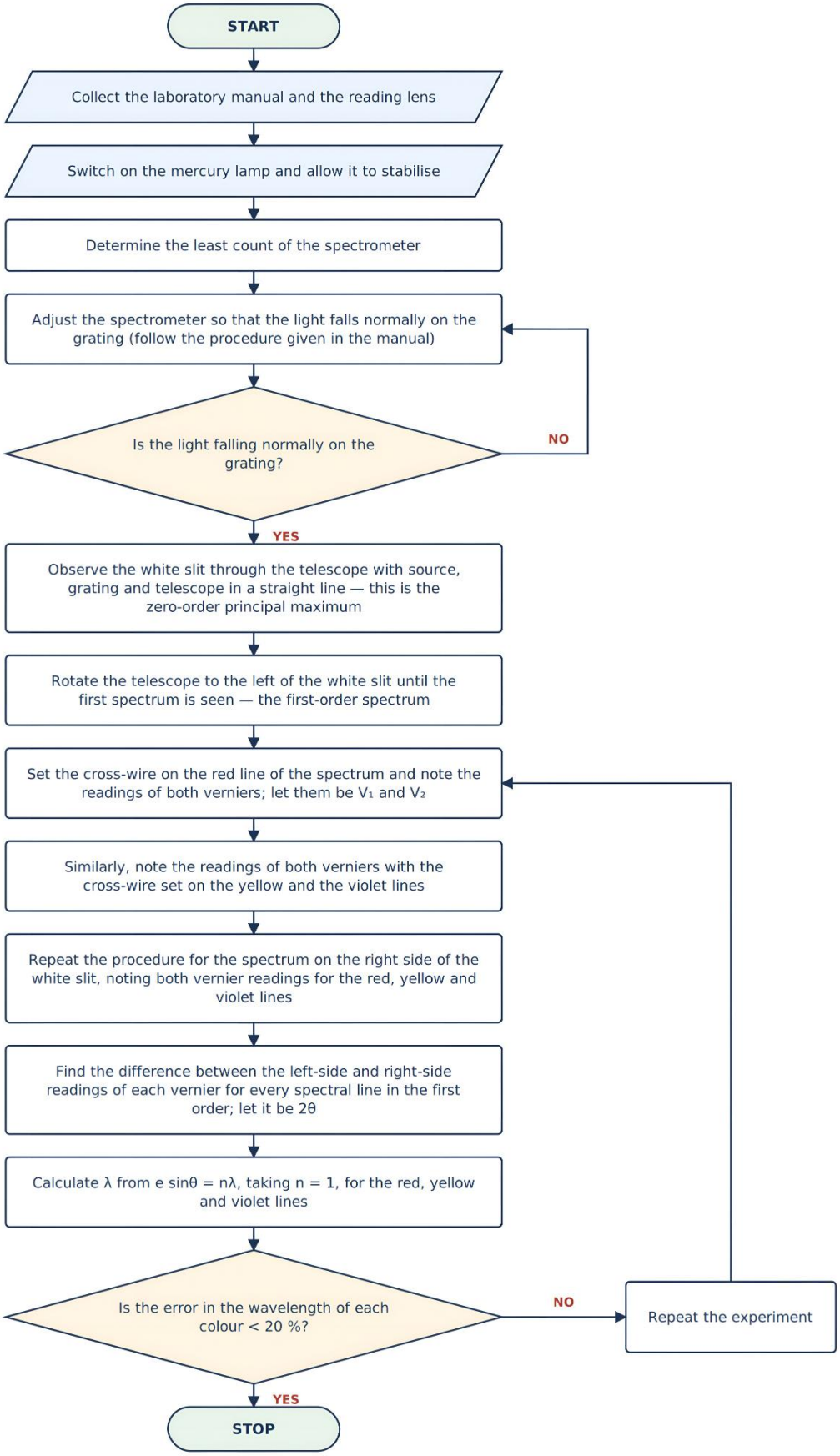
Percentage error:

$$\% \text{ error} = \frac{Std\ Value - Expt\ Value}{StdValue} \times 100 =$$

5.8. Precautionary measures to be taken:

1. Both the telescope and the collimator should be focused for parallel rays.
2. The eyepiece should be focused on the cross-wires.
3. The Prism table should be properly leveled.
4. While taking observations, slit should be made narrow and the cross-wires as a multiplication mark should be adjusted at the center of the image.

Exp. 5. Flow Chart



5.9. Applications of Spectrometer Experiment

1. Calibration of spectrometers.
2. Identification of unknown elements.
3. Study of atomic spectra.
4. Determination of energy levels.
5. Wavelength calibration of optical instruments.

5.10. Some sample viva-voice questions:

1. What is the principle of experiment and its condition to see pattern
2. by bare eyes?
3. What is Grating element?
4. What is Diffraction Grating?
5. Define essential parts of spectrometer.
6. What is the main function of diffraction grating?
7. Instead of mercury source if we use sodium vapor lamp, what changes can we observe?
8. What do you mean by diffraction?
9. 1 inch = …….. cm.
10. What is the lease count of a spectrometer?
11. Why does red deviate the most in case of grating?

MELDE'S EXPERIMENT

To determine the frequency of an electrically maintained tuning fork by Melde's experiment.

7.1. Objective: To determine the frequency of an electrically maintained tuning fork by Melde's experiment.

7.2. Apparatus/Components Required:

i) Electrically maintained tuning fork
ii) Lightweight pan
iii) Weight box
iv) Power supply
v) Lightweight string
vi) Stand with clamp and pulley

7.3. Theory: This experiment is based on resonance between an exciter of waves, the tuning fork and the medium that supports the wave, i.e., the stretched string. The tuning fork has its own natural frequency that depends upon its physical dimensions and the elasticity of the material used for its fabrication. This is constant for a particular tuning fork.

When one end of a thin, long, flexible string is attached to one of the arms of the tuning fork, while the other end is subjected to tension, the string resonates only at frequencies given by

$$f = \frac{p}{2l}\sqrt{\frac{T}{m}}$$

where,

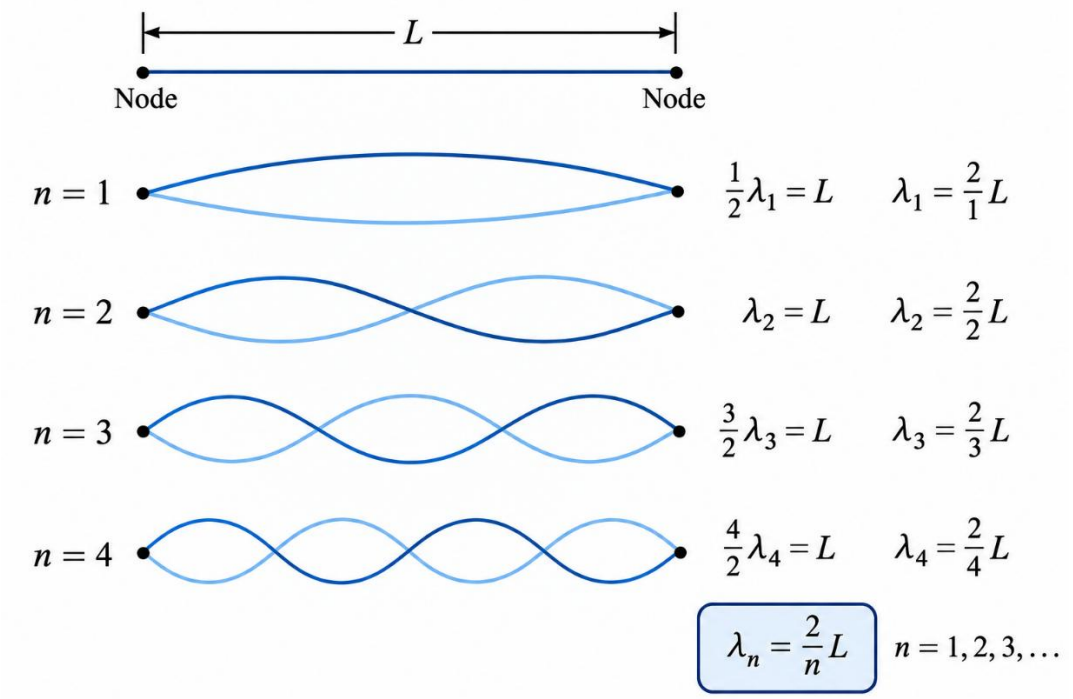
T is the tension in the string

(T=M × g= ($mass\ of\ pan + mass\ on\ the\ pan$) × g),

m is its mass per unit length or the linear density of the string,

l is the length of the stretched portion of the string

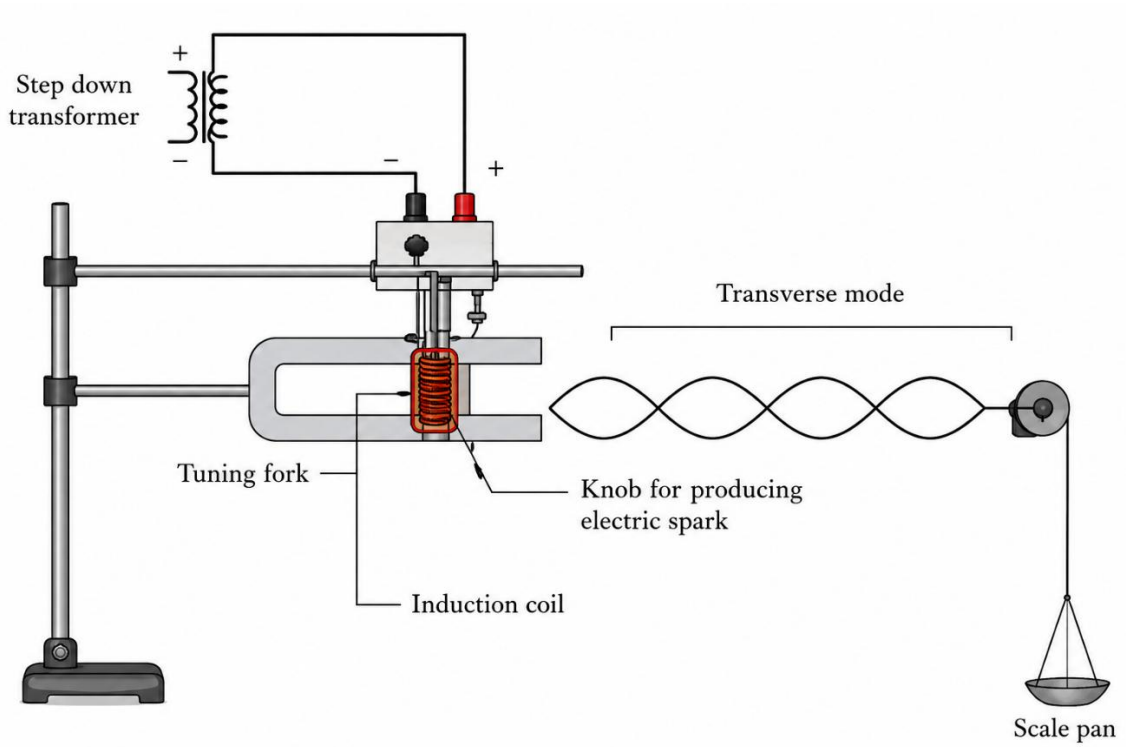
The natural frequency of the string depends on its length. If the length of the stretched portion of the string is changed to match the frequency of the tuning fork, the string resonates and well-defined stationary loops are formed.



7.1 Figure: Schematic representation of standing waves showing nodes and antinodes.

The waves on the string can be excited in two different ways: (i) by tuning fork vibrations across the length of the string, or (ii) by tuning fork vibrations along the length of the string. In the first case, the frequency of the exciter (tuning fork) is the same as that of the waves on the string. In the second case, during one complete oscillation of the exciter, the string undergoes only half a cycle; hence, the frequency of the tuning fork is twice the frequency of the wave on the string.

Principle of working: The experiment works on the principle of resonance of a stretched string driven by an electrically maintained tuning fork. When the frequency of the stretched string matches the driving frequency condition, a standing-wave pattern with well-defined nodes and antinodes is formed. By measuring the resonating length, counting the number of loops, and determining the tension in the string, the frequency of the tuning fork can be calculated.



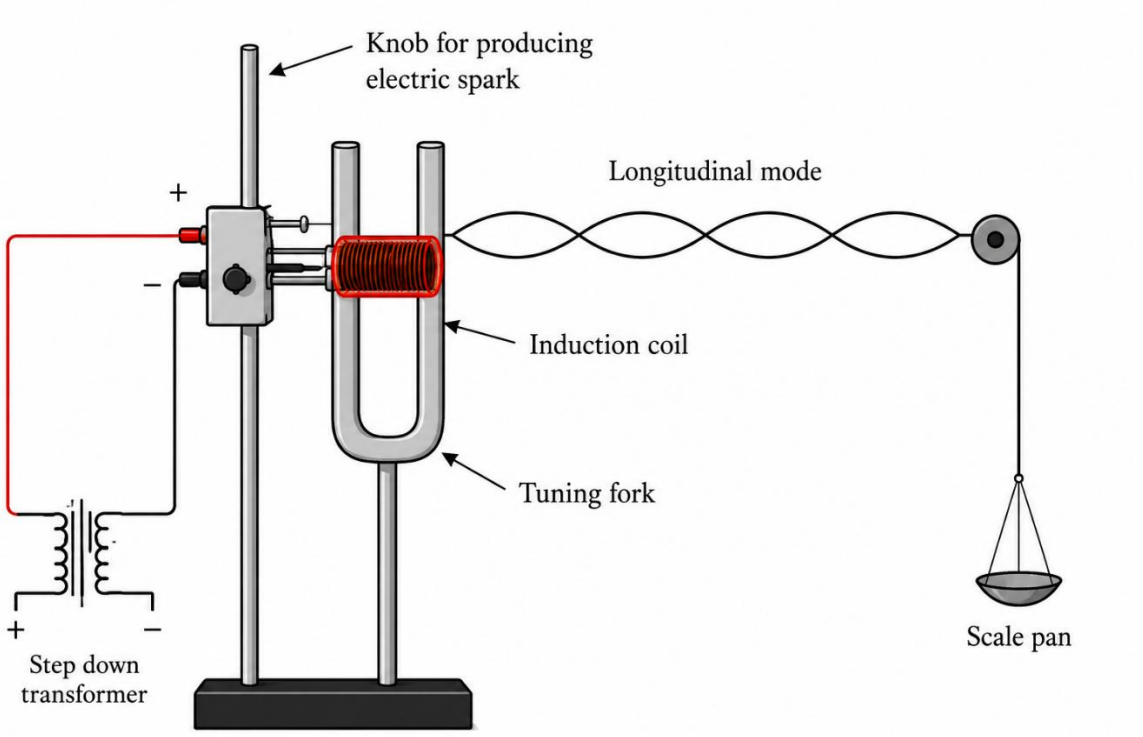
7.2 Figure: Experimental setup for the determination of the frequency of the tuning fork using resonance in the attached string. Here, the tuning fork vibrates across the length of the string.

When the tuning fork vibrates across the length of the string, the string completes one vibration along with the tuning fork. Hence, in this mode, the frequency of the tuning fork is equal to the frequency of the string

$$f = \frac{p}{2l}\sqrt{\frac{T}{m}}$$

$$f = \frac{p}{2l}\sqrt{\frac{Mg}{m}}$$

where $T\ (= M \times g)$ is the tension in the string, M is the total mass of the pan and the weights placed on the pan, m is the mass per unit length of the string, l is the length of the string between its two fixed ends, and p loops are formed in length.



7.3 Figure: Experimental setup for the determination of the frequency of the tuning fork using resonance in the attached string. Here, the tuning fork vibrates along the length of the string.

When the tuning fork vibrates along the length of the string, the string completes only one-half of the vibration while the tuning fork completes one.

Hence, in this mode, the frequency of the tuning fork is double the frequency of the string,

$$f = \frac{p}{l}\sqrt{\frac{T}{m}}$$

$$f = \frac{p}{l}\sqrt{\frac{Mg}{m}}$$

where $T\ (= M \times g)$ is the tension in the string, M is the total mass of the pan and the weights placed on the pan, m is the mass per unit length of the string, l is the length of the string between its two fixed ends, and p loops are formed in length.

7.4. Procedure:

1. Arrange the apparatus as shown in the figure.
2. Excite the tuning fork by switching on the power supply. An operating voltage between 6 V and 8 V is advisable. Adjust the screw provided for this purpose to obtain sustained vibrations.
3. Once the tuning fork vibrations are steady, make the attached string pass over the pulley and put some load on the pan attached to its other end.
4. The pulley is fixed on a stand at one end of the table. The tuning fork is stationed on a heavy base. Shift the tuning fork with its base, either towards the pulley or away from it gently, to alter the length so that well-defined loops are formed sustainably.
5. Note the length of the string and the corresponding number of antinodes. Record all relevant data in the observation table.
6. Change the load on the pan and repeat the process.

7.5. Observations

Mass per unit length of the string (m) =…….. gm/cm

Mass of pan = …….. gm; g = 980 cm/sec^2

Observation table:

S. No.	Mass applied (gm)	Mass of Pan (gm)	Total Mass(M)	Tension (M x g)	No. of loops (p)	Total effective length L (in cm)	Frequency (n)
1							
2							
3							
4							
5							
						$\bar{f}$=	

7.6. Calculation:

In longitudinal mode

$$f = \frac{p}{l}\sqrt{\frac{Mg}{m}}$$

7.7. Results:

The experimentally determined frequency of the tuning fork when vibrating along the length of the string is……..Hz.

Percentage error:

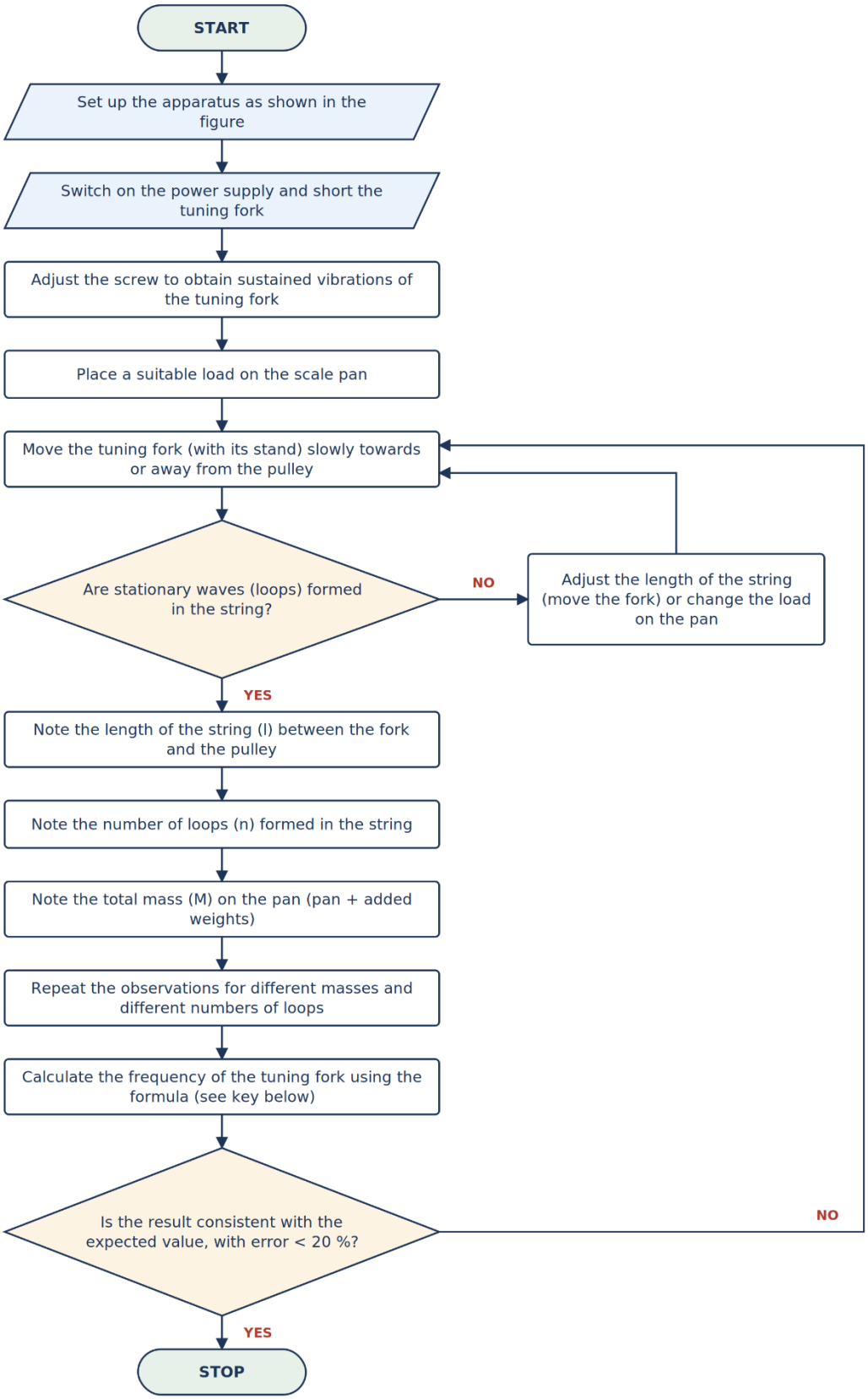
$$\% \text{ error} = \frac{\text{Std Value} - \text{Expt Value}}{\text{StdValue}} \times 100\% = \ldots\ldots\ldots\ldots\%$$

Standard value for the given tuning fork is in between 75 to 100 Hz.

7.8. Precautionary measures to be taken:

1. Friction in the pulley should be minimized as far as possible.
2. The string should be light, flexible and properly stretched.
3. The tuning fork should be operated at a suitable voltage; very high voltage should be avoided.
4. The load should be added or removed carefully without disturbing the setup.
5. The tuning fork should be shifted gently while adjusting the resonating length.
6. Clear and stable loops should be obtained before taking readings.
7. The length of the resonating portion of the string should be measured carefully.
8. The number of loops should be counted accurately.
9. Vibrations and accidental movement of the apparatus should be avoided.
10. Ensure that the pulley rotates freely and the string passes smoothly over it.

Exp. 7. Flow Chart



7.9. Applications:

1. Determination of the frequency of an unknown tuning fork or other periodic source using resonance.
2. Study of the relation between frequency, tension, length and linear density of a stretched string.
3. Demonstration of resonance and formation of stationary waves in a stretched string.
4. Measurement and study of the properties of strings used in musical instruments.
5. Educational demonstration of standing waves, nodes, antinodes and resonance.

7.10. Some sample viva-voce questions:

Q1. What is the principle of Melde's experiment?
Q2. What is meant by resonance in a stretched string?
Q3. What is the role of the tuning fork in this experiment?
Q4. What is meant by linear density of the string?
Q5. How is the tension in the string determined?
Q6. What are nodes and antinodes?
Q7. What happens when the tension in the string is increased?
Q8. What happens to the resonating length when the tension is changed?
Q9. Why should friction in the pulley be minimized?
Q10. Why is the tuning fork electrically maintained?
Q11. What is the difference between transverse and longitudinal modes in Melde's experiment?

Q12. Why is the tuning fork frequency twice the string frequency in the longitudinal mode?

Q13. Why should very high voltage not be applied to the tuning fork?

Q14. How can the accuracy of the experiment be improved?

PLANCK'S CONSTANT

Determination of Planck's constant (h) using light emitting diode (LED) of various colors.

6.1. Objective: Determination of Planck's constant (h) using light emitting diode (LED) of various colors.

6.2. Apparatus/Components Required:

1. Planck's constant apparatus
2. connecting wires

6.3. Theory: The Planck's constant (h) is given by Max Planck to explain the concept of fundamental unit of energy, a quantum, to explain the spectral distribution of black body radiation. In order to explain the black body radiation, Planck proposed that atoms absorb and emit radiation in discrete quantities given by-

$$E = n\,hf,$$

Where n is an integer known as quantum number, ν is the frequency, h *is* Planck's constant. In applications where frequency is expressed in terms of radians per second ("angular frequency") instead of cycles per second, it is often useful to absorb a factor of 2π into the Planck constant. The resulting constant is called the reduced Planck constant or Dirac constant. It is equal to the Planck constant divided by 2π, and is denoted $\hbar$ ("h-bar"):

$$\hbar = h/2\pi$$

Band-gap energy E_g:

$$E_g = \frac{hc}{\lambda}$$

The energy of a photon with angular frequency? where $w = 2\pi v$, is given by $E = \hbar w$

In this experiment, we will use light emitting diode (LED) which is semiconductor device. This device emits radiations in visible and Infra-Red (IR) frequencies when a suitable voltage is applied to them. The symbol of LED as shown in Figure 6.1:

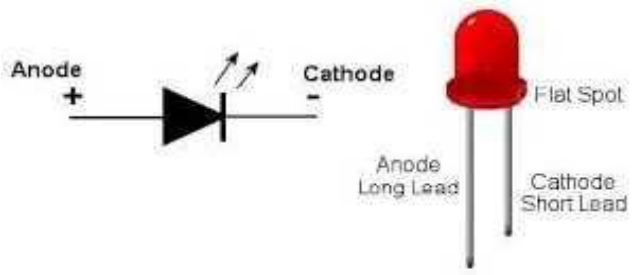


Figure 6.1: LED and symbol

LED emits light only when the voltage is forward biased and above a minimum threshold value. These combinations of condition create an electron hole pair in a diode. Electron

hole pairs are charge carriers and move when placed in an electrical potential. Thus, many electron hole pairs are producing current when placed in an electric field. The commercially used LED's have a typical voltage
drop between 1.5 Volt to 2.5 Volt or current between 10 to 50 mA. The exact voltage drop depends on the LED current, color, tolerance, and so on.

When LED is forward-biased, the current is carried (mostly) by electrons in the higher energy conduction band of the N type material and by holes moving in the opposite direction in the lower energy p-material. At the p-n junction, electrons meet and "fall into" the conduction band to the valence band, giving up a quantum of energy in an LED type diode; this energy is emitted as a photon in the visible or infrared spectrum. The training board circuit diagram is shown in Figure 6.2.

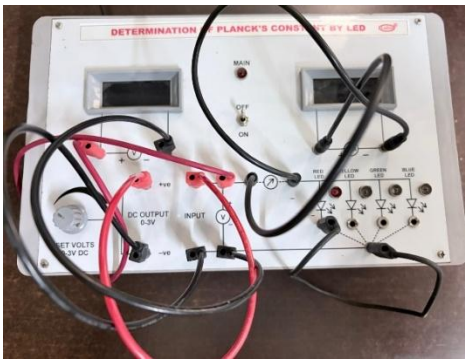


Figure 6.2: Training board diagram

Each photon of light is emitted with energy equal to the energy difference between the bands. In application, the light emitted can be used as a substitute for a small incandescent bulb, for example as an indicator light on an instrument, or as part of a digital display. The color (frequency) of the light emitted by an LED is determined during its manufacture by making a choice among materials with various band energies. A semiconductor diode requires a certain minimum forward voltage (threshold voltage) to initiate significant current flow and light emission.

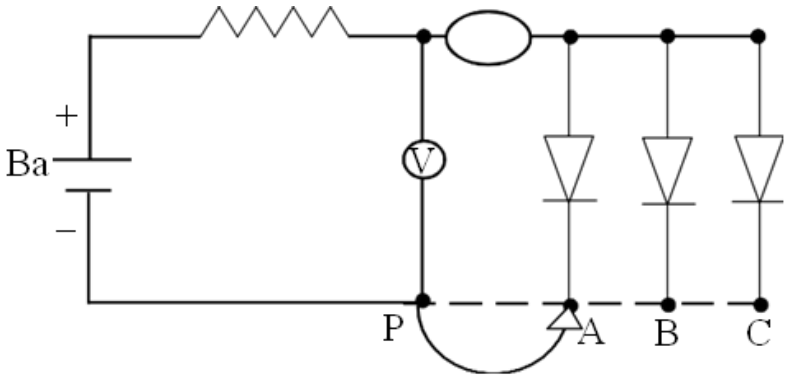


Figure 6.3: Circuit diagram of forward biasing of LED

Each photon of light is emitted with energy equal to the energy difference between the bands. In application, the light emitted can be used as a substitute for a small incandescent bulb, for example as an indicator light on an instrument, or as part of a digital display. The color (frequency) of the light emitted by an LED is determined during its manufacture by making a choice among materials with various band energies. A semiconductor diode requires a certain minimum forward bias voltage before it will conduct much current. If it is an LED, at least this minimum voltage is required before it will light up.

As explained later; this "turn on" voltage (V) is related to the Planck's constant as:

$$hf = eV + \Delta \qquad (1)$$

where Δ is current from one LED to another of the same type (which is very less so neglected), e is the electric charge, eV represents the energy received from the power supply by each electron crossing the junction.

We use several LED's that emit light of different colors (frequencies). We measure the turn-on voltage for each and make a graph of frequency (f) vs Energy (eV) according to Eq. (1). The slope of the graph is Planck's constant (h).

6.4. Procedure:

1. Make the connections as shown in Figure 6.3 for Red LED.
2. Connect the voltmeter, ammeter and preferred LED at appropriate place in the circuit.
3. Vary the DC supply voltage and note down corresponding voltmeter and ammeter readings.
4. Plot the graph of voltage V (on X axis) versus current I (on Y axis) for given LED and draw tangent to the curve the point of intersection of tangent with X axis gives the value of turn on voltage. The turn on Voltage is the minimum voltage required for glow of LED.
5. Repeat the same procedure for other LEDs (Yellow, green and violet).
6. Calculate the frequency (f) and Energy (E) using formula $f = c/\lambda$ and $E = eV$, respectively.
7. Substitute the value of frequency and Energy in Table.
8. Calculate the Planck's constant using h= E/f

6.5. Observations: For the determination of Planck's constant using LED of various colour.

Observation table:

S. No.	RED LED		YELLOW LED		GREEN LED	
	Voltage (in V)	Current (mA)	Voltage (in V)	Current (mA)	Voltage (in V)	Current (mA)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Plot the graph between voltage V (on X axis) vs current I (on Y axis) for all color LED and calculate the Turn on Voltage by drawing tangent to the curve for each color LED. The point of intersection of tangent and X axis gives you the value of turn on voltage.

The Turn "ON" voltages of:

- Red LED …… volts,
- Yellow LED ……. volts,
- Green LED ……. Volts
- Blue LED ……. Volts

S. No.	Color of LED	λ (Å)	f=C/λ in (Hz)	Turn on Voltage V(volt)	E in (Joules) (eV)	Planck's constant h= E/f (Joule sec)
1	Red					
2	Yellow					
3	Green					
4	Blue					

Standard Value:

Planck's constant is 6.6 x10^{-34} Js

$c = 3.00 \times 10^{8}$ m/s

$e = 1.6021765 \times 10^{-19}$ coulomb

Wavelength: Red - 6600Å

Yellow - 6100Å

Blue - 4700Å

Green - 5250Å

6.6. Calculation:

$$\boldsymbol{E = hf, h = \frac{E}{f}}$$

$$\boldsymbol{E = eV_0 \text{ and } f = \frac{C}{\lambda}}$$

Where, V_0 = Turn on Voltage (Find out from Graph)

6.7. Results:

The Planck's constant is found to be ……..

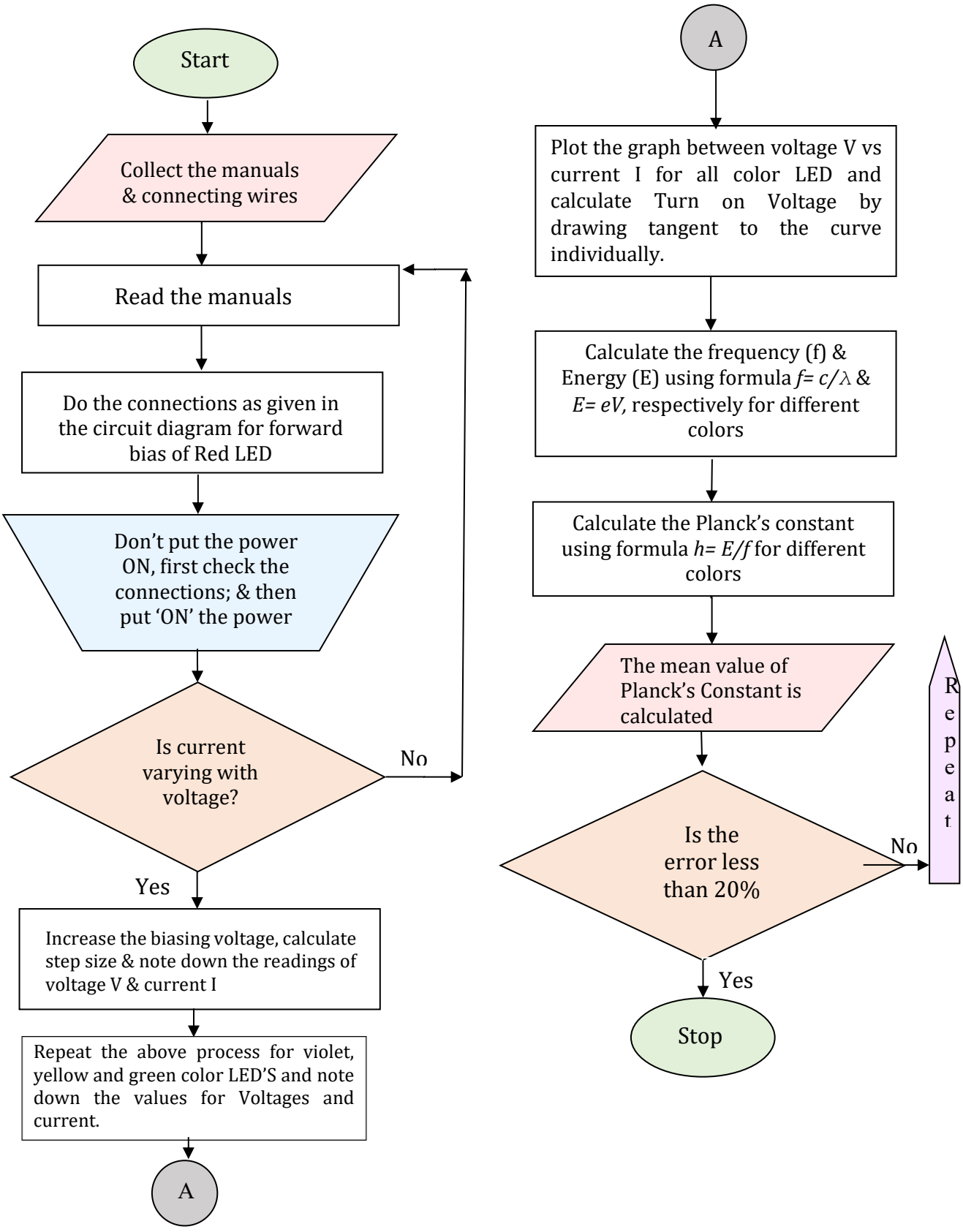
Percentage error:

$$\% \text{ error} = \frac{\text{Std Value} - \text{Expt Value}}{\text{StdValue}} \times 100 =$$

6.8. Precautionary measure to be taken:

1. Connections should made be properly and tightly.
2. At least five readings should be taken

Exp. 6. Planck's Constant Flow Chart



6.9. Applications of Planck's Constant experiment

1. Verification of Quantum Theory:

It experimentally verifies the quantization of energy and the relation between photon energy and frequency:

$$E = h\vartheta$$

2. Determination of Photon Energy:

Planck's constant is used to calculate the energy of photons of different frequencies or wavelengths.

3. Study of LED Characteristics:

The experiment helps determine the turn-on voltage and relate it to the energy band gap of the semiconductor.

4. Determination of Semiconductor Band Gap:

For an LED, the approximate band-gap energy can be obtained from:
and hence the relationship between band gap and emitted wavelength can be studied.

5. Understanding Electroluminescence:

It explains how electrical energy is converted into light energy through electron-hole recombination in LEDs.

6.10. Some sample viva-voice questions:

i. What is Planck's constant?
ii. What is LED and its working principle?
iii. Which color shows lowest turn on voltage and why?
iv. Why are LEDs of different colors used?
v. Why does an LED emit light when forward biased?
vi. Why does the color of an LED depend on its material?
vii. Why is a resistor connected in series with the LED?
viii. How is band-gap energy related to the LED wavelength?
ix. What happens to photon energy when wavelength decreases?
x. What is threshold voltage?

GM COUNTER

To study the variation of count rate with applied voltage of a Geiger-Müller (GM) counter, determine its operating voltage, and estimate the efficiency of the GM detector for a radioactive source.

8.1. Objective:

To study the variation of count rate with applied voltage of a Geiger-Müller (GM) counter, determine its operating voltage, and estimate the efficiency of the GM detector for a radioactive source.

8.2. Apparatus Required:

Geiger-Müller Counter Setup**,** GM Tube, connecting cables and mains power cord**,** Radioactive Source (e.g., ^{60}Co, β^-)**,** Source Holder/Stand with adjustable distance

8.3. Theory and working principle:

The Geiger-Müller (GM) counter, developed by Hans Geiger and Walther Müller, detects ionizing radiation (α, β, or γ particles) based on **gas ionization and Townsend avalanche multiplication**. The detector consists of a hollow metallic cylinder acting as a cathode, filled with a low-pressure inert gas (typically argon or neon) mixed with a quenching agent. A fine wire running along the tube's central axis serves as an anode, maintained at a high positive potential ($+400$ to $+850$ V). When an incident ionizing particle penetrates the tube window, it ionizes gas atoms to create primary electron-ion pairs.

Driven by the strong electric field, the primary electrons accelerate rapidly toward the central anode wire. As they gain kinetic energy, they collide with neighboring gas molecules, liberating additional secondary electrons in a cascading chain reaction known as a **Townsend avalanche**. This massive charge amplification generates a single, strong electrical discharge per incoming particle, regardless of its initial energy. The resulting current pulse produces a voltage drop across an external resistor, which an electronic

circuit amplifies and registers as a count. The added quenching gas neutralizes positive ions to prevent secondary false discharges, quickly resetting the detector for subsequent radiation events.

When ionizing radiation enters the tube, it ionizes gas molecules, creating ion pairs.

$$\text{Radiation} + \text{Ar} \rightarrow \text{Ar}^{+} + e^{-}$$

A **Townsend avalanche** (or Townsend discharge) is a gas ionization process critical to electrical breakdown and radiation detectors. If the electron gains enough energy before colliding with neutral gas molecules, impact ionization occurs, releasing an additional electron. These resulting electrons accelerate and cause further ionizations, creating a rapidly multiplying cascade. This exponential "avalanche" dramatically increases electrical conduction, playing a vital role in gaseous detectors like Geiger-Müller tubes.

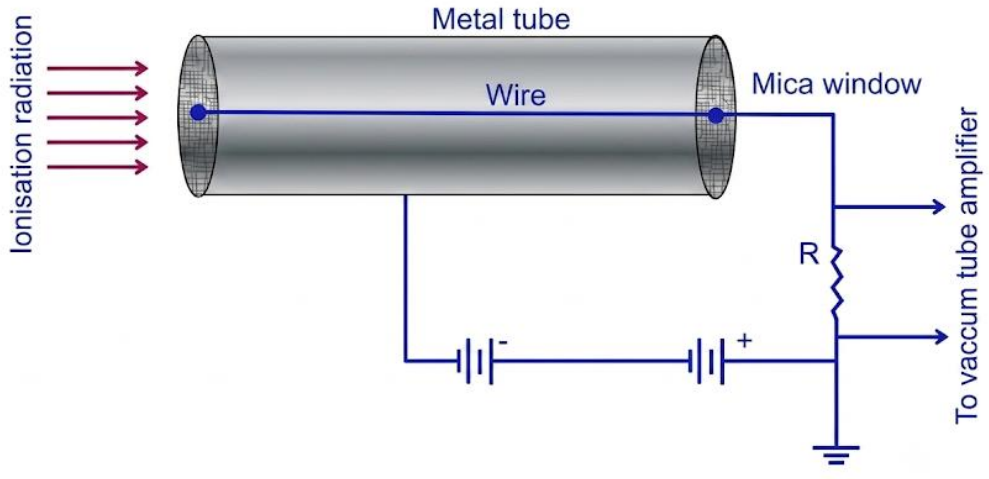


Figure 8.1(a): Block Diagram of GM Counter

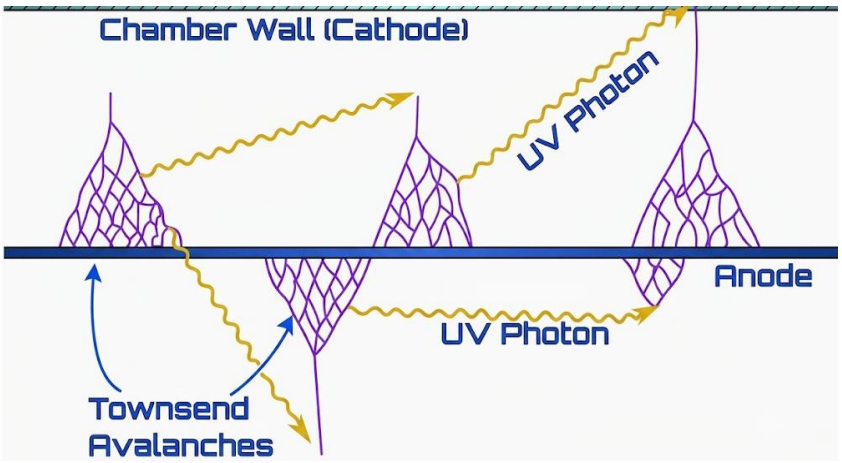


Figure 8.1(b): Townsend avalanche

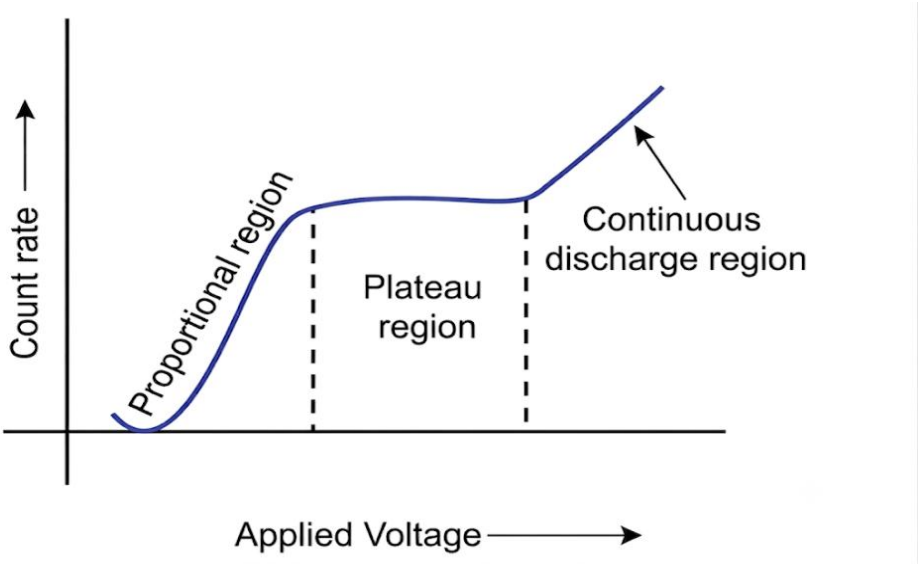


Figure 8.2: GM Characteristics Curve

Starting Voltage: The minimum voltage at which pulses are first detected.

Plateau Region: A range of voltage where the count rate remains relatively constant despite voltage increases. This is the practical working limit of the tube.

Breakdown Region: Beyond the plateau, where the tube enters continuous discharge, tube can be damaged.

Operating Voltage & Plateau Slope: The **Operating Voltage** is typically chosen at the midpoint of the plateau to ensure stable performance. The **Plateau Slope** indicates the quality of the tube; a lower slope (usually 2-3%) is preferred.

8.4. Procedure:

1. Set voltage to starting threshold (350-400 V) without a source. Record background counts over a fixed time interval in 50 V steps up to 850 V.
2. Place the radioactive source on a designated shelf in the sample holder beneath the tube window. Reset voltage to 350-400 V and repeat count measurements in 50 V steps up to 850 V over the same time interval.
3. Subtract background counts from total counts at each voltage step:

$$\text{Real Counts} = \text{Total Counts} - \text{Background Counts}$$

4. Plot Voltage (x-axis) vs. Real Counts (y-axis). Find the flat plateau region and calculate its midpoint:

$$V_{\text{op}} = \frac{V_1 + V_2}{2}$$

5. Set the counter to V_{op}, measure the net count rate, and compute intrinsic efficiency using the manual's geometry and activity formula.
6. Turn the high-voltage knob down to zero before switching off main power to prevent voltage surges.

8.5. Observations

Observation table (Set time for 30 Second)

S. No.	Voltage (V)	Count without sample (N_b)	Count with sample (N_s)	Real count ($N_s - N_b$)

Graph Plot:

Plot a graph of **Real Count** on the Y-axis vs. **Applied Voltage (V)** on the X-axis.

8.6. Calculations:

Operating Voltage:

$\boldsymbol{V_{operating}} = \frac{\boldsymbol{V_1}+\boldsymbol{V_2}}{2}$ *(where* $\boldsymbol{V_1}$ *and* $\boldsymbol{V_2}$ *are the start and end voltages of the plateau).*

The starting voltage of the GM tube was found to be ……..V.

The plateau region of the GM counter exists from $\boldsymbol{V_1}$= …….. V to $\boldsymbol{V_2}$= ……..V.

The operating voltage of the GM counter was determined to be $\boldsymbol{V_{operating}}$ =

……V

Efficiency of the GM counter:

Radioactive Source used = ……..

Current Activity of the radioactive source; A = …….. kBq (Refer Note)

Distance from source holder to end window; D = …… (Refer Note)

Diameter of end window d = 0.9 cm

N_S = Counts recorded with source for 30 sec =……..

N_b= Counts recorded due to background for 30 sec = ……

CPS = Real count rate recorded per second; $N = \frac{N_S - N_b}{100}$ = ……

This source is radiating isotopically in all directions. A fraction of this only is entering the GM tube per second is calculated as; $R = \frac{A \times d^2}{16\, D^2} =$ ……

Intrinsic Efficiency of the GM counter

$$\eta = \frac{\text{Measured Count Rate (CPS)}_{net}}{\text{Actual Activity of Source (DPS)}_{window}} \times 100\ \% = \frac{N}{R} \times 100\ \% =$$ …… %

8.7. Results:

The operating voltage of the GM counter $\boldsymbol{V_{operating}}$ = …… V

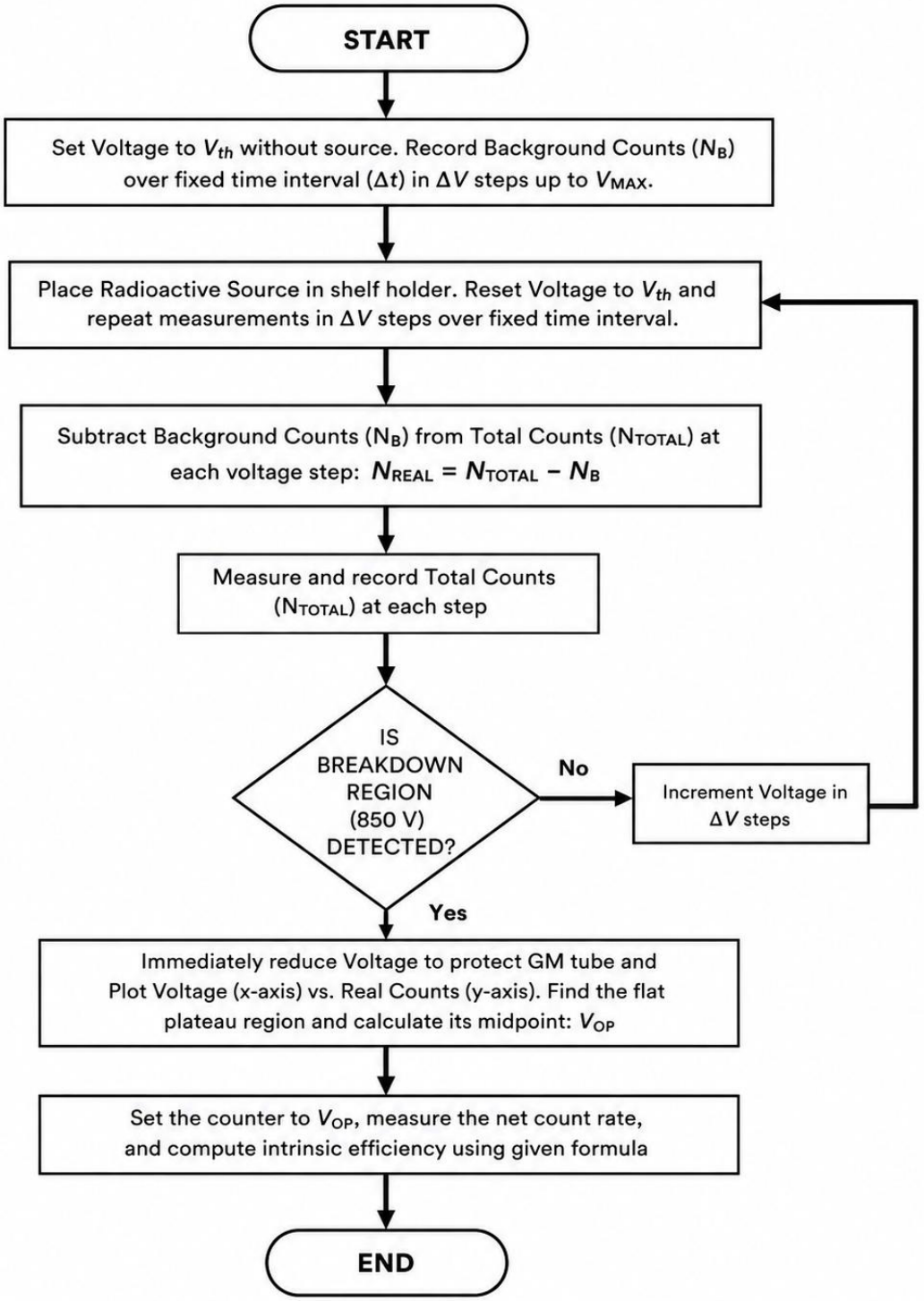
Intrinsic Efficiency of the GM counter $\eta =$ ……

8.8. Precautionary measure to be taken:

1. Do not touch the mica window of the GM tube as it is extremely thin and fragile.
2. Never exceed the plateau voltage beyond 850 V; continuous discharge in the breakdown region will permanently damage the tube.
3. Handle radioactive sources with care. Don't touch in bare hand to the center of samples.
4. Ensure count rates do not exceed 100,000 counts/min to avoid significant dead-time (coincidence) errors.
5. Keep mobile phones and other electronic devices away from the detector to prevent false signals.

6. Amplifiers and detectors are always running at high voltage. So please don't switch off any part during the data acquisition, it may damage the amplifier and detector.

Exp. 8. Flow Chart



8.9. Applications for GM Counter experiment

- **Environmental Radiation & Safety Monitoring:** Detects background radiation levels and identifies surface contamination in nuclear power facilities, research laboratories, and environmental survey sites.
- **Medical Physics & Healthcare Dosimetry:** Measures stray radiation exposure in radiology and nuclear medicine departments to ensure dose limits are safe for healthcare personnel and patients.
- **Industrial Quality Control & Non-Destructive Testing:** Utilizes count rate changes to measure material thickness (such as paper, glass, or metal sheets) and detect leaks in sealed pipelines.
- **Geological Prospecting & Mining:** Helps locate deposits of naturally occurring radioactive materials (NORM), such as uranium and thorium ores, during field exploration.
- **Nuclear Physics Research & Education:** Used to determine radioactive half-lives, test the effectiveness of different shielding materials, and calculate detector operating efficiency.

Several used terms for GM Counter

Dead Time (t_d)

Immediately after an ionization event, a dense sheath of slow-moving positive ions forms around the central anode wire. This temporary "space charge" drops the electric field strength so low that the tube cannot trigger another avalanche, even if a new particle enters.

Example: If a GM tube has a dead time of 200 μs and a second cosmic ray enter 100 μs after the first, it will go completely undetected.

Recovery Time (t_r)

As the positive ion cloud drifts away toward the outer cathode, the electric field slowly recovers. The recovery time is the interval required for the field to return to its original strength, allowing full-sized pulses to be detected again.

Paralysis Time / Resolving Time (τ)

The resolving time is the minimum time interval between two distinct counts. It encompasses the dead time plus recovery time needed for a pulse to cross the counter's electronic detection threshold.

Example: If $\tau = 300\ \mu s$ and the true count rate is extremely high, the tube stays continuously "paralyzed" in a discharged state, and the registered count rate might paradoxically drop to zero.

Quenching

When positive ions hit the cathode, they can release secondary electrons, causing an endless, false loop of continuous discharging. **Quenching** is the process of stopping this. It is achieved chemically by adding a halogen gas (like chlorine) that absorbs the energy without releasing electrons, or electronically by momentarily dropping the high voltage.

Note: If your source is old, you must first calculate its current attenuated activity using the radioactive decay law $A = A_0 e^{-\lambda t}$

For Example: ^{60}Co Source in the LAB

Initial Activity (A_0): 37 kBq (in November 2024)

Time Elapsed (t): 2 years (from November 2024 to November 2026)

Half-life of Cobalt-60 ($T_{1/2}$): 5.27 years (standard physical constant)

Current Activity (A): 37 kBq × 0.7687 ≈ 28.44 kBq (in November 2026)

Sr. No.	Source Name	Current Activity A (kBq)	Half Life ($T_{1/2}$) in years	D (cm)
1	^{60}Co	28.44	5.27	2
2	^{137}Cs	53.62	30.17	6

HEISENBERG UNCERTAINITY PRINCIPLE

To demonstrate the Heisenberg Uncertainty Principle using single-slit diffraction with a laser.

9.1 Objective:

To demonstrate the Heisenberg Uncertainty Principle using single-slit diffraction with a laser.

9.2 Apparatus Required:

Semiconductor Laser, Single slit, Optical bench, Photodetector and Meter scale

9.3 Theory: When monochromatic light passes through a narrow slit whose width is comparable to its wavelength, the light bends around the edges of the slit. This phenomenon is known as diffraction

According to the Huygens–Fresnel principle, every point on the slit acts as a source of secondary wavelets. The interference of these wavelets produces a diffraction pattern consisting of a bright central maximum surrounded by alternating dark and bright fringes.

The condition for minima is

$$a \sin\theta = n\lambda \text{ ------- (9.1)}$$

Where a is the width of single slit

n is the order of minima

λ is the wavelength of Laser source

From Eq. (9.1) we can calculate the width of the slit as

$$a = \frac{n\lambda}{\sin\theta}$$

Heisenberg Uncertainty Principle

The Heisenberg Uncertainty Principle states that it is impossible to determine simultaneously the exact position and momentum of a particle.

Mathematically, it is given as

$$\Delta y . \Delta p_x \geq \frac{h}{4л} \text{ ..... (9.2)}$$

This expression implies that when a monochromatic wave travel to a single slit, it has a definite momentum p_x, in x-direction before it is getting into the slit, but we are not exact about its position

When the beam hits at the slit, we can define an uncertainty in position for the beam in the vertical direction in the order of slit width 'a' (because we do not know where the photons strike in a vertical slit of width 'a') we can define it to be equal to the slit width

$$\Delta y = a$$

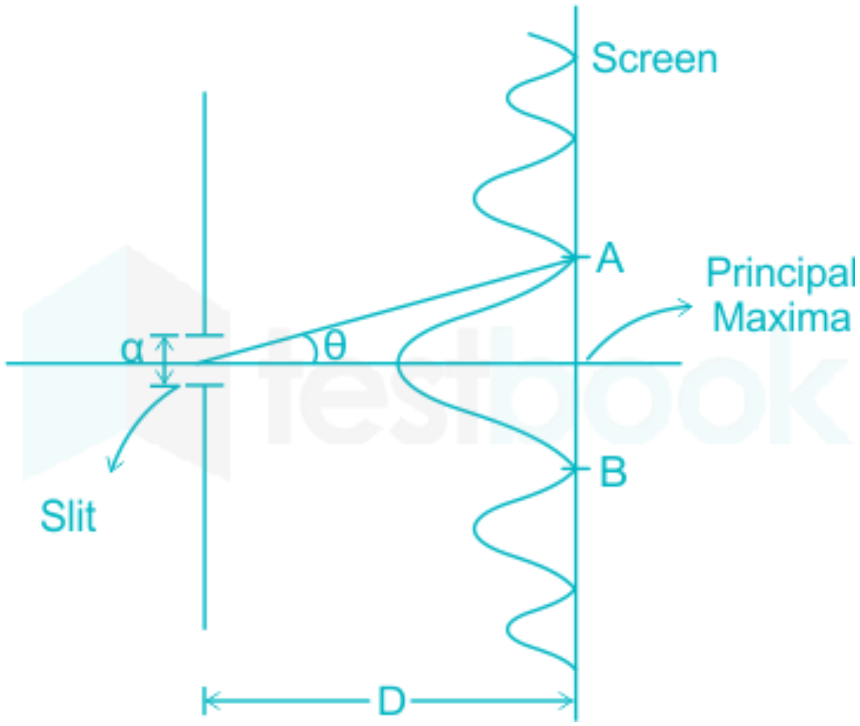


Figure 9.1, Intensity profile of monochromatic light due to single slit diffraction

We know that as the beam passes through the slit, diffraction pattern is produced on the screen, from this it is clear that as photons travel through a

single slit, it acquires a momentum in y-direction p_y. It is because we know that some of the photons go straight and other fall at an angle 'θ' from the normal to the screen and using trigonometry relation, the uncertainty in the momentum can be represented as

$$\Delta p_y = p.\sin\theta \ldots\ldots\ldots\ldots\ldots (9.3)$$

From the de-broglie hypothesis

$$p = \frac{h}{\lambda} \quad \ldots\ldots\ldots (9.4)$$

$$\text{and} \quad a\sin\theta = n\lambda$$

Now, substitute the value of p and $\sin\theta$ for first minima in eq (9.3), we get

$$\Delta p_y.\Delta y = h \ldots\ldots\ldots\ldots\ldots (9.5)$$

By this equation, we can verify Hiesenberg's Uncertainty Principle.

9.4 Procedure:

1. Switch ON the Laser supply and Output measurement unit.
2. Couple the laser light into the narrow hole of the detector to get maximum output current.
3. Insert the slit into the cell mount between detector and source and focus the light on the slit.
4. Move the detector in step (x) on the extreme left or right to obtain max output current starting from zero current.
5. Plot the graph between distance moved by detector y (mm) and output current I (mA).
6. Measure the position of first minima from the diffraction pattern with the help of graph.
7. Calculate the value of diffraction angle 'θ' using formula $\theta = y/D$.

8. Find the slit width using formula $a = \frac{\lambda}{sin\theta}$.
9. Calculate the uncertainty in the momentum using $\Delta p_y = \frac{h}{\lambda} \cdot \sin\theta$.
10. Now verify Heisenberg Uncertainty Principle.

9.5 Observations:

To find slit width

λ = 655 nm

Pitch of Micrometre 0.5 mm and L.C = 0.01 mm

Observation Table:

S. No	Micrometre reading (mm)	Output Current(mA)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		

Plot the graph between Distance (y) vs Current (I). Find the position for first minima. The position of first minima is y…….. (mm).

9.6. Calculation:

To find slit width

Distance between detector and slit D=……..

The position of first minima y =……..

Angle made by first minima $\theta = \tan^{-1}\left[\frac{y}{D}\right]$ =……..

The slit width is $a = \frac{m\lambda}{\sin\theta}$ =……..

For verification of the result,

$$\Delta y = a$$

From the de-Broglie hypothesis

$$p = \frac{h}{\lambda}$$

The uncertainty of momentum is

$$\Delta p_y = p.\sin\theta$$

$$\Delta p_y = \frac{h}{\lambda}\sin\theta$$

$$=\ldots\ldots$$

We apply the Heisenberg Uncertainty Principle, according to the principle

$$\Delta y.\Delta p_y = h.$$

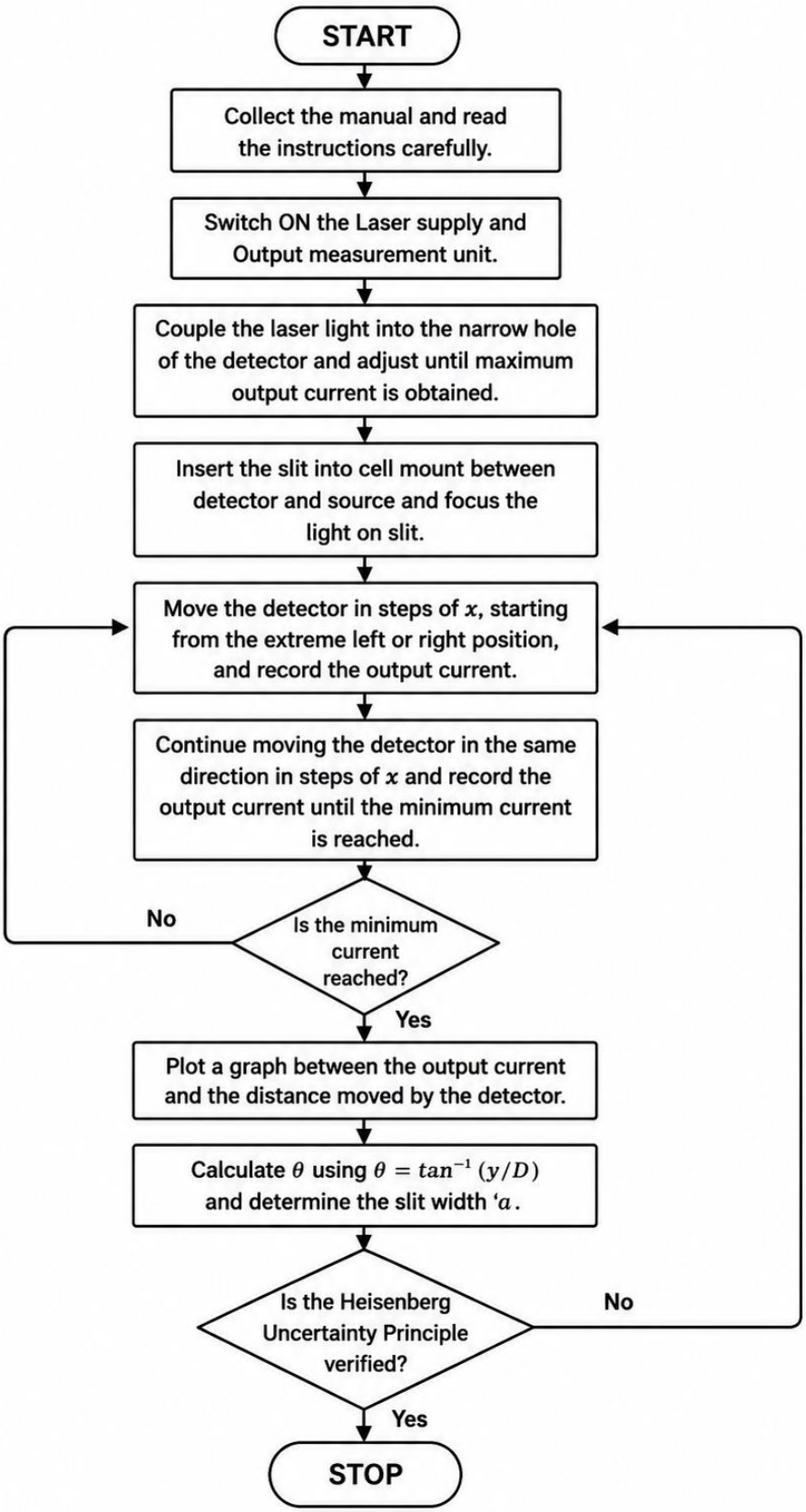
9.7. Results: Check whether Heisenberg's uncertainty principle is verified or not.

9.8 Precautions:

1. The laser beam should be properly aligned with the slit.
2. Avoid direct exposure of the laser beam to the eyes.
3. The screen should be perpendicular to the laser beam.

4. Measurements should be taken carefully to minimize parallax error.
5. The laboratory should be free from vibrations and air disturbances.
6. Record observations only after obtaining a sharp diffraction pattern.

Exp. 9. Flow Chart:



9.10. Some sample viva-voice questions:

1. What is Heisenberg's Uncertainty Principle?
2. State the mathematical form of the uncertainty principle.
3. What do you mean by position uncertainty (Δx)?
4. What do you mean by momentum uncertainty (Δp)?
5. Is the uncertainty principle applicable only to microscopic particles?
6. Why is the uncertainty principle a fundamental principle of quantum mechanics?
7. Can a particle have simultaneously definite position and momentum?
8. Why does a narrow slit cause a larger spread in the diffraction pattern?

HALL EFFECT

To study the Hall Effect experiment and find the Hall coefficient and carrier concentration of a given semiconductor crystal.

10.1. Objective: - To study the Hall Effect experiment and find the Hall coefficient and carrier concentration of a given semiconductor crystal.

10.2. Apparatus/Components Required: - Hall probe, Hall Effect set-up, Electromagnet, constant current power supply, digital Gauss meter.

10.3. Theory:

When a sample (metal or semiconductor) carrying a current I placed in a transverse magnetic field B, then an electric field E is induced in the direction perpendicular to both I and B. This phenomenon is called Hall Effect.

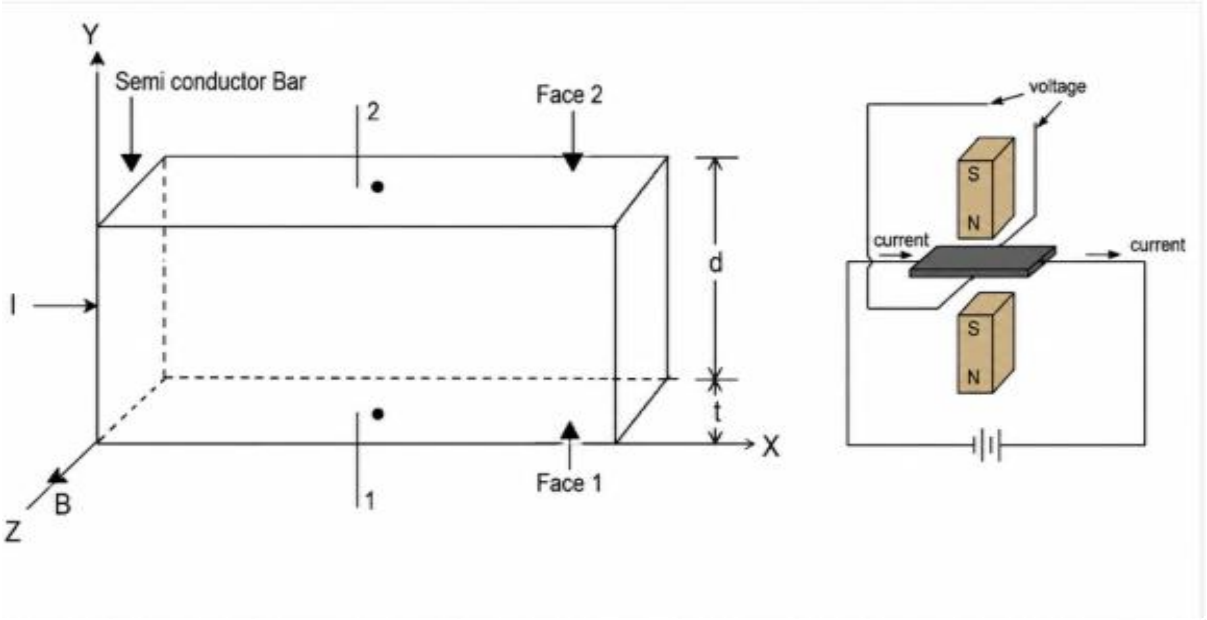


Figure 10.1: Schematic diagram illustrating the Hall effect in a semiconductor bar and the experimental arrangement for measuring the Hall voltage.

Figure 10.1 shows a semiconductor bar carrying a current I in the positive X direction. Let a magnetic field B is applied in the positive Z direction. Now a force is exerted on the charge carriers (electrons or holes) in the negative Y direction. Due to this force, charge carriers are pressed downwards toward face 1. For example, in n type sample, the charge carriers are electrons which are accumulated on face 1. So, face 1 becomes negatively charged with respect to face

2. Therefore, a potential difference V_h is developed between surface 1 and 2 which is called the Hall voltage. The polarity of Hall voltage enables us to determine experimentally whether the semiconductor sample is *n type or p type*.

In equilibrium state, the electric field intensity E due to Hall Effect must exert a force on the carriers which just balances the magnetic force i.e.,

$$eE = Bev \qquad (10.1)$$

Where e is the magnitude of charge on electron or hole and v is the drift velocity

Now,

$$E = \frac{V_h}{d} \text{ or } V_h = E \times d$$

$$\therefore V_h = Bvd \qquad [\because E = Bv] \qquad (10.2)$$

We know that the current density J is given by

$$J = \frac{I}{A} = nev$$

Where I is current, v is the velocity, e is electronic charge and $A = t\,.\,d$ is the area of the sample.

$$\therefore v = \frac{I}{neA} \qquad (10.3)$$

From equations (10.2) and (10.3), we have

$$V_h = B \times \frac{I}{neA} \times d = \frac{B \times I}{ne \times t}$$

Now we introduce Hall coefficient $\boldsymbol{R_H}$ defined by

$$\boldsymbol{R_H = \frac{1}{ne} = \frac{V_h \times t}{B \times I}} \qquad (10.4)$$

Carrier concentration $(n) = \frac{1}{neR_H}$ (10.5)

Connection with your experimental graph:

In your experiment, you plot: $V_h\ vs.\ B$

From $V_h = \frac{R_H \times I}{t} \cdot B$

This has the form $y = m\,x$

Therefore, the slope of the $V_h\ vs.\ B$ graph is

$$Slope = \frac{\Delta V_h}{\Delta B} = \frac{R_H \times I}{t}$$

Hence,

$$R_H = slope \times \frac{t}{I}$$

Then the carrier concentration is

$$n = \frac{1}{e|R_H|}$$

10.4. Procedure:

Determination of Hall Voltage

EXPERIMENTAL PROCEDURE:

1. Connect the green wires to the current terminals and the red wires to the voltage terminals of the Hall Effect setup.
2. Set the semiconductor sample current (I) to a fixed value, preferably $I >= 4\ mA$. Keep the magnetic field OFF and measure the corresponding ohmic voltage (V_O).
3. Place the semiconductor crystal exactly perpendicular to the magnetic field and symmetrically between the poles of the electromagnets.
4. Switch ON the electromagnet and increase the magnetic current (I_h) in steps of $0.25\ A$. For each value of magnetic current, note the corresponding magnetic field (B) and measured voltage (V_m).
5. Take six consecutive values of magnetic current, for example: $I_h = 0.25, 0.50, 0.75, 1.00, 1.25\ and\ 1.50\ A$. Record the corresponding values of magnetic field (B) and measured voltage (V_m) in the observation table.
6. Calculate the Hall voltage (V_h) for each magnetic current value using: $V_h = V_m - V_O$
7. Plot Hall voltage (V_h) on the Y-axis and magnetic field (B) on the X-axis.
8. Draw the best-fit straight line through the experimental points and calculate its slope using:

$$Slope = \frac{\Delta V_h}{\Delta B}$$

9. Calculate the Hall coefficient (R_H) using:

   $R_H = \frac{Slope \times t}{I}$ where, t = thickness of the semiconductor sample and I = current through the semiconductor sample

10. Calculate the carrier concentration (n) using: Carrier concentration (n) = $\frac{1}{neR_H}$ where, e = electronic charge and R_H = Hall coefficient.

11. Finally, calculate the percentage error to find out the deviation of calculated values of the Hall coefficient (R_H) and carrier concentration (n) from the standard value.

10.5. Observations

Specification of the sample:

Crystal: $p\ type$ lightly doped/$n\ type$ lightly doped…….

Thickness of the material: ……..

Semiconductor Current ($\geq\ 4mA$) :……

Observation table:

Sample current (mA)	***Ohmic Voltage Vo (mV) (Magnetic Flux is zero)***	**Magnetic current (A)**	**Magnetic Flux B (Gauss)**	**Measured Voltage *Vm (mV)***	**Hall Voltage (mv)** $V_h = V_m - V_o$

10.6 Calculation:

Plot graph between **Magnetic Field (B) (X-axis). vs. Hall Voltage (V_h) (Y-axis)**

Calculate the slope

$$\textbf{Slope (S)} = \frac{\boldsymbol{\Delta V_h}}{\boldsymbol{\Delta B}} =$$

The Hall Coefficient

$$R_H = \frac{S \times t}{I}$$

where t is the thickness of the sample, and I is the sample current flowing through the sample.

The Hall coefficient $\boldsymbol{R_H}$ = ……

10.7 Result: -

The Hall Effect set up is studied and the various parameters measured for the give crystal are as follows:

Hall coefficient $\boldsymbol{R_H}$ =

Carrier concentration n (cm^{-3}) =

Standard value:-

For $p-$ $type$ semiconductor

$$\boldsymbol{R_H} = 38.86 \times 10^{-3}\ \boldsymbol{m^2/Coulomb}$$

Carrier concentration = $1.60 \times 10^{20} cm^{-3}$

For $n-$ $type$ semiconductor

$$\boldsymbol{R_H} = 24.77 \times 10^{-3}\ \boldsymbol{m^2/Coulomb}$$

Carrier concentration = $2.52 \times 10^{20} cm^{-3}$

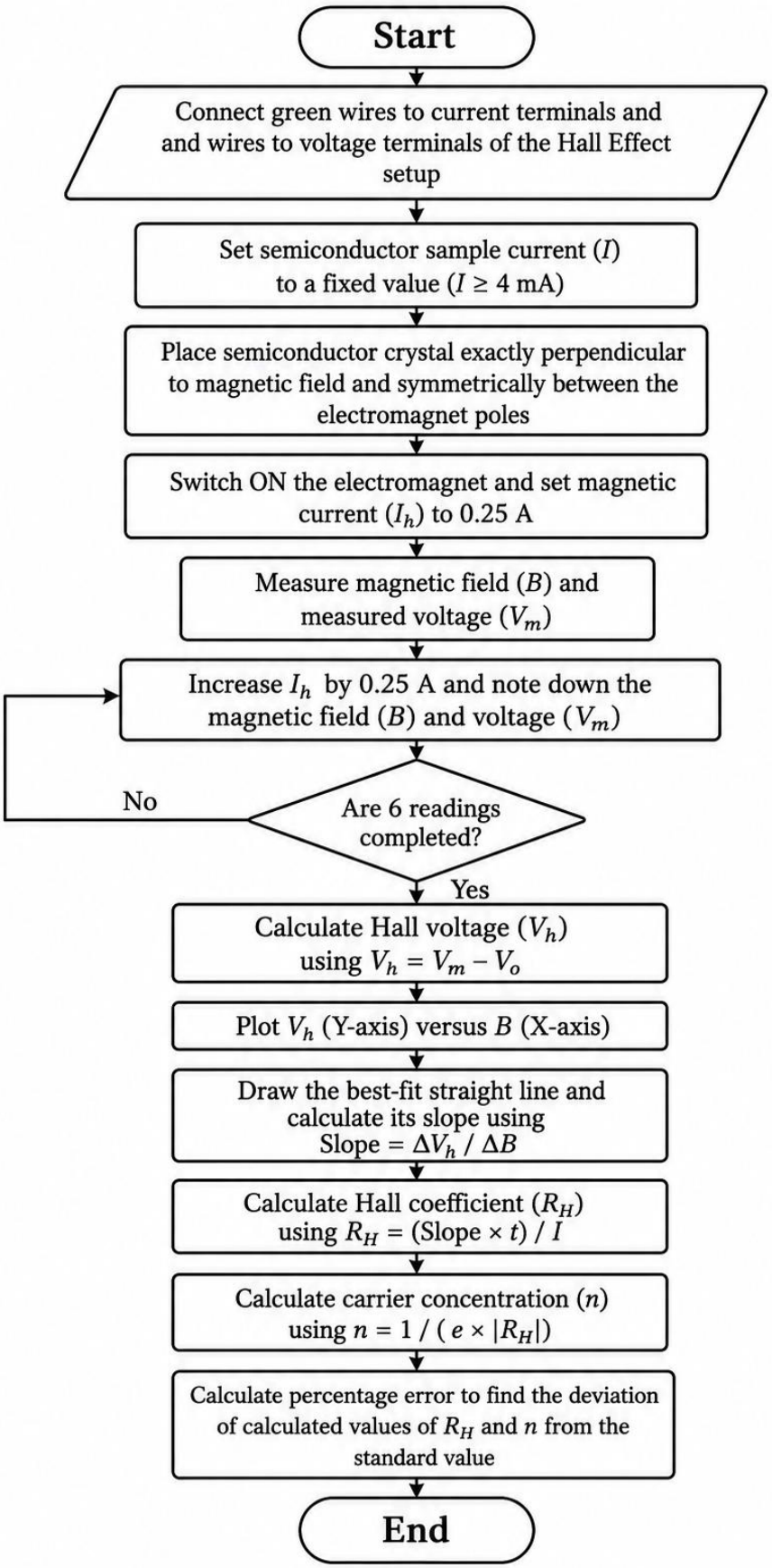
Percentage error:

$$\%\ \text{error} = \frac{Std\ Value - Expt\ Value}{StdValue} \times 100 =$$

10.8 Precautions:

1. While noting the value of Hall voltage the directions of current and magnetic field should be changed simultaneously.
2. The zero fields potential should be as small as possible and should be subtracted from the measured potential.
3. Hall probe should be handled very carefully.
4. Readings should be noted carefully.

Exp. 10. Flowchart:



10.9. Applications of Hall Effect

1. Determination of semiconductor type – Identifies whether a semiconductor is n-type or p-type from the polarity of the Hall voltage.
2. Measurement of carrier concentration – The Hall coefficient can be used to determine the number density of charge carriers in a semiconductor.
3. Determination of carrier mobility – Hall measurements help calculate the mobility of charge carriers.
4. Measurement of magnetic field – Hall sensors are widely used to measure the strength of magnetic fields.
5. Current sensing – Hall-effect sensors can measure electric current without directly connecting to the current-carrying conductor.
6. Position and speed sensing – Used in motors, automobiles, fans, and industrial systems to detect the position or rotational speed of moving components.

10.10. Some sample viva-voice questions:

1. What is the Hall Effect?
2. What is Hall voltage?
3. What is the Hall coefficient?
4. What is the SI unit of Hall coefficient?
5. Why is the Hall Effect more prominent in semiconductors?
6. Why does Hall voltage develop?
7. What is the Lorentz force?
8. What is the direction of Hall voltage?
9. How can you determine whether a semiconductor is n-type or p-type?
10. What is the sign of Hall coefficient for an n-type semiconductor?
11. What is the sign of Hall coefficient for a p-type semiconductor?
12. How does Hall voltage vary with current?
13. How does Hall voltage vary with the magnetic field?
14. What happens to Hall voltage when the magnetic field is reversed?
15. What happens when the direction of current is reversed?
16. Why is the semiconductor sample placed perpendicular to the magnetic field?
17. Why are four contacts used on the semiconductor sample?

18. Why is the ohmic voltage measured before applying the magnetic field?
19. What is the purpose of the electromagnet in this experiment?
20. Why should the semiconductor sample be placed symmetrically between the magnet poles?
21. What happens if the semiconductor sample is not properly aligned?
22. How is carrier concentration determined from the Hall coefficient?
23. How is carrier mobility determined?
24. What is the relation between Hall coefficient and carrier concentration?
25. What are the practical applications of the Hall Effect?
26. What is the difference between Hall voltage and ohmic voltage?
27. Why is a constant current maintained through the semiconductor?
28. What is meant by charge-carrier concentration?
29. What are the charge carriers in n-type and p-type semiconductors?
30. What are the major sources of error in the Hall Effect experiment?

Virtual Lab web link

1. Introduction: For calculation of Least Count.
   a. Vernier Calliper: https://www.olabs.edu.in/?sub=1&brch=5&sim=16&cnt=4
   b. Screw Gauge: https://www.olabs.edu.in/?sub=1&brch=5&sim=156&cnt=4
   c. Spectrometer: https://ov-amrt.vlabs.ac.in/exp/diffraction-grating/theory.html

2. Cyclotron: https://virtuelle-experimente.de/en/b-feld/anwendung/zyklotron.php

3. Numerical Aperture: https://lo-amrt.vlabs.ac.in/exp/numerical-aperture-optical-fiber/index.html
   (https://hmw-amrt.vlabs.ac.in/exp/meldes-string-apparatus/simulation/flash/flashplayer.exe)
   (https://hmw-amrt.vlabs.ac.in/exp/meldes-string-apparatus/simulation/flash/simulation/index.swf)

4. Newton's Ring: https://lo-amrt.vlabs.ac.in/exp/newtons-rings-wavelength-light/index.html

5. Wavelength using grating: https://ov-amrt.vlabs.ac.in/exp/diffraction-grating/theory.html

6. Planck's Constant: https://mp-amrt.vlabs.ac.in/exp/determination-plancks-constant/index.html

7. Melde's Exp.: https://hmw-amrt.vlabs.ac.in/exp/meldes-string-apparatus/index.html

8. Hall Effect: https://ssp-amrt.vlabs.ac.in/exp/hall-effect-charge-carrier/index.html